

FAULT DETECTION AND CLASSIFICATION IN POWER SYSTEMS

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CERTIFICATION

This is to certify that the research project titled **“FAULT DETECTION AND CLASSIFICATION IN POWER SYSTEMS”** submitted to the Department of Systems Engineering, University of Lagos, is a record of the original project construction carried out by **DAUDU, USMAN ADEKUNLE (170407011)** for the award of Bachelor of Science (BSc. Hons.) in Systems Engineering, University of Lagos, Akoka, Lagos.

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DEDICATION

This project is dedicated to God Almighty, my ever-supportive and relentless parents and all entities in the education system working assiduously to pass on knowledge and skills needed to continue the growth of mankind and make the world a better place to be.

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I wish to register my profound gratitude to Almighty God for his guidance and mercy throughout my life.

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ABSTRACT

Faults in power systems are the occurrence of a significant deviation from the normal operational values in electrical quantities such as currents, voltages and frequencies. We say that a fault has occurred when there is a significant change in the current, voltage or frequency values of power output beyond the operational values which can damage the equipment of consumers or cause a power outage.

The aim of this project is to develop a procedure which requires minimal time for the detection of faults in the transmission system and is also robust enough to work accurately in a dynamic and diverse range of electrical networks to quickly detect faults and implement protective measures ensuring the safety of connected electrical components.

A novel approach was developed which makes use of the moving averages and moving variances of the voltages measured across transmission lines for real-time detection of faults which was validated using a MATLAB SIMULINK model of the IEEE 9 bus transmission system. The slowest fault detection time observed using the approach presented in this project is 3.50 milliseconds while the fastest detection time took 0.67 milliseconds with the majority of faults being detected between 1.00 milliseconds and 0.67 milliseconds.

In conclusion, the findings from this research show that by using moving averages and moving variances of voltages in a transmission system, faults can be detected faster which would result in increased sustainability and reliability of the electrical grid.

LIST OF FIGURES

Figure 1.1: Structure of an Electrical Power System. Source: (Sulaiman, 2017).....	1
Figure 1.2: Line-to-Line-to-Line (L-L-L) Fault	2
Figure 1.3: Line-to-Line-to-Line-Ground (L-L-L-G) Fault	3
Figure 1.4: Single line-to-ground (L-G) faults.....	3
Figure 1.5: Line-to-Line (L-L) Faults.....	4
Figure 1.6: Line-to-Line-to-Ground (L-L-G) Faults	4
Figure 1.7: Effects of Various Faults on Voltage and Current. Source: (YUMURTACI et al., 2016).....	5
Figure 2.1: Fahmida and Jorge's Approach. Source: (Fahmida N. Chowdhury & Jorge L. Aravena, 1998).....	10
Figure 2.2: Module I of Fahmida and Jorge's Approach. Source: (Fahmida N. Chowdhury & Jorge L. Aravena, 1998)	11
Figure 2.3: Neural Network Architecture in Fahmida and Jorge's Approach. Source: (Fahmida N. Chowdhury & Jorge L. Aravena, 1998).....	11
Figure 2.4: Flowchart of Approach Proposed by Aida Asadi Majd and his co-authors. Source: (Aida Asadi Majd et al., 2017)	13
Figure 2.5: Architecture of a Wide Area Measurement System (WAMS). Source: (Halil Alper Tokel et al., 2018).....	14
Figure 2.6: Delay in Data Arrival. Source: (Halil Alper Tokel et al., 2018)	14
Figure 2.7: Visualization of the Approach of Gustavo Santos and his co-authors. Source: (Gustavo Santos et al., 2019).....	15
Figure 2.8: Processes of Fault Detection, Classification and Location. Source: (Radu-Adrian Tirnovan & Maria Cristea, 2019).....	16
Figure 2.9: Three Processes Involved in the Approach of Nakho and his co-authors. Source: (Nakho et al., 2021)	19
Figure 2.10: Flowchart for the Approach of Ragab Mahmoud (Part I). Source: (Ragab Mahmoud, 2022)	20
Figure 2.11: Flowchart for the Approach of Ragab Mahmoud (Part II). Source: (Ragab Mahmoud, 2022)	21
Figure 2.12: Processes Involved in the Approach of Pavan Venkata and his co-authors. Source: (Pavan Venkata et al., 2022).....	22

Figure 2.13: Processes Involved in the Approach of Ehab M. Esmail et al. Source: (Ehab M. Esmail et al., 2023)	24
Figure 3.1: Effect of Fault on Current	31
Figure 3.2: Effect of Fault on Voltage	32
Figure 3.3: Fault Detection Using Moving Variance	33
Figure 3.4: Fault Detection Using Moving Average.....	34
Figure 3.5: IEEE 9 Bus Transmission System Downloaded from MATLAB Central.....	35
Figure 3.6: Splitting of the PI Sections (Transmission Lines).....	36
Figure 3.7: Addition of Three-Phase Circuit Breaker	37
Figure 3.8: Modified IEEE 9 Bus Transmission System Used for Research.....	38
Figure 3.9: Dashboard Showing Information about Fault	39
Figure 3.10: Fault Detection Subsystem in Simulink.....	42
Figure 3.11: Fault Detection Process for each Bus	43
Figure 3.12: Process of Calculating Moving Statistics.....	43
Figure 3.13: Fault Detection Using the Moving Statistics	44
Figure 3.14: Fault Line Identification Subsystem	44
Figure 3.15: Fault Type Identification	45
Figure 3.16: Fault Location Subsystem	45
Figure 3.17: Simulation Powergui	46
Figure 3.18: Block Diagram showing the processes in the proposed approach	47
Figure 4.1: Fault Occurrence and Detection of AB Fault on Line 7-8 at 50km from Bus 7 ..	48
Figure 4.2: Performance of Fault Transmission Line Identification Model	53
Figure 4.3: Performance of Fault Type Identification Model.....	54
Figure 4.4: Relay System Activation.....	55
Figure 5.1: Count of Fault Detection Times	56
Figure 5.2: Distribution of Fault Detection Time	57
Figure 5.3: Distribution of Fault Detection Times Based on Transmission Line of Occurrence	58
Figure 5.4: Distribution of Fault Detection Times Based on Type of Fault	59

LIST OF TABLES

Table 2.1: Features Used in Approach Presented by Abdulaziz Aljohani. Source: (Abdulaziz Aljohani, 2020)	18
Table 2.2: Summary of Literatures.....	30
Table 3.1: Features used in the Approach Presented in this Project.....	41
Table 4.1: Fault Detection Times Per Type of Fault on Transmission Line 7-8	49
Table 4.2: Fault Detection Times Per Type of Fault on Transmission Line 8-9	50
Table 4.3: Fault Detection Times Per Type of Fault on Transmission Line 9-6	50
Table 4.4: Fault Detection Times Per Type of Fault on Transmission Line 6-4	51
Table 4.5: Fault Detection Times Per Type of Fault on Transmission Line 4-5	52
Table 4.6: Fault Detection Times Per Type of Fault on Transmission Line 5-7	52
Table 5.1: Statistics from the Distribution of Fault Detection Time	57

TABLE OF CONTENTS

CERTIFICATION.....	i
DEDICATION	ii
ACKNOWLEDGEMENT	iii
ABSTRACT	iv
LIST OF FIGURES.....	v
LIST OF TABLES.....	vii
TABLE OF CONTENTS	viii
CHAPTER ONE: BACKGROUND AND MOTIVATION.....	1
1.1 Introduction	1
1.1.1 Symmetrical Faults	2
1.1.2 Unsymmetrical Faults	3
1.2 Problem Statement	6
1.3 Aim of the Study.....	7
1.3.1 Objectives of the Study.....	7
1.4 Justification of Study	7
1.5 Significance of the Study.....	8
1.6 Scope and Limitation.....	8
CHAPTER TWO: REVIEW OF PAST WORK.....	10
2.1 Literature Review	10
2.2 Summary of Literatures.....	25
2.3 Contribution to Research	30
CHAPTER THREE: METHODOLOGY	31
3.1 Intuition behind Fault Detection.....	31
3.2 Voltages as Parameters for Fault Detection.....	32
3.3 Fault Indicator	32
3.3.1 Moving Variance.....	33

3.3.2	Moving Average.....	33
3.4	Data Generation.....	34
3.4.1	Modified IEEE 9 Bus Transmission System.....	36
3.4.2	Data Collection	39
3.5	Fault Detection	42
3.6	Fault Line Identification	44
3.7	Fault Type Identification	44
3.8	Fault Location.....	45
3.9	Relay System	45
3.10	Block Diagram.....	47
CHAPTER FOUR: RESULTS AND ANALYSIS		48
4.1	Introduction	48
4.2	Fault Detection	48
4.2.1	Fault Detection on Line 7-8.....	49
4.2.2	Fault Detection on Line 8-9.....	49
4.2.3	Fault Detection on Line 9-6.....	50
4.2.4	Fault Detection on Line 6-4.....	51
4.2.5	Fault Detection on Line 4-5.....	51
4.2.6	Fault Detection on Line 5-7.....	52
4.2.7	Overall Slowest and Fastest Fault Detection Time	53
4.3	Transmission Line Identification	53
4.4	Fault Type Identification	53
4.5	Fault Location.....	54
4.6	Relay System	55
CHAPTER FIVE: SUMMARY, CONCLUSION AND RECOMMENDATION.....		56
5.1	Summary.....	56
5.1.1	Overall Distribution of Detection Times	56

5.1.2	Distribution of Detection Time Based on Line of Fault.....	58
5.1.3	Distribution of Detection Time Based on Type of Fault.....	59
5.2	Conclusion	60
5.3	Recommendation	61
REFERENCES.....		62
APPENDIX A: DATASET GENERATION CODE		65
APPENDIX B: FULL DATASET GENERATION CODE		68
APPENDIX C: CODE FOR BUILDING LINE IDENTIFICATION MODEL		69
APPENDIX D: CODE FOR BUILDING FAULT TYPE IDENTIFICATION MODEL.		70
APPENDIX E: CODE FOR BUILDING FAULT LOCATION MODEL		71
APPENDIX F: CODE FOR GETTING FAULT DETECTION TIMES		72

CHAPTER ONE: BACKGROUND AND MOTIVATION

1.1 Introduction

The electrical power system is often referred to as the electrical grid. It consists of a large interconnection of electrical components forming the backbone of modern civilization. This vast network is comprised of three major systems which are:

- i. the generation systems which include the power plants.
- ii. the transmission systems which include step-up transformers, transmission lines, and step-down transformers.
- iii. and the distribution systems which include substations, and distribution transformers.

A typical structure of the power system is shown in figure 1.1 below.

Electricity generation, transmission, and distribution

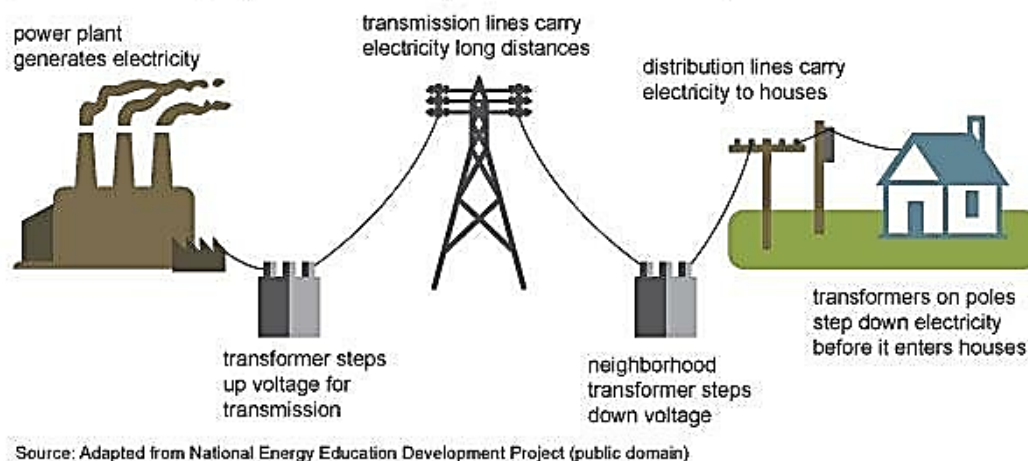


Figure 1.1: Structure of an Electrical Power System. Source: (Sulaiman, 2017)

Continuity of service and reliability is a vital attribute of an electrical power system (Qais Alsafasfeh et al., 2010). This attribute requires an optimal balance between the supply of electricity being generated at the power plant and the demand from the consumers which involves the transmission of electricity through long distances along the transmission lines and distribution networks. The transmission line (followed by the distribution systems) is the most exposed and vulnerable part of the power system to atmospheric phenomena and damage from human activity (Julio Barros & José M. Drake, 1994). Apart from not having enough power generation to meet demand, this vulnerability has made the majority of interruptions to the

continuous flow of electricity to be resulted from faults along the transmission lines or distribution networks (Soufiane Belagoune et al., 2021).

Faults in power systems are the occurrence of a significant deviation from the normal operational values in electrical quantities such as currents, voltages and frequencies. We say that a fault has occurred when there is a significant change in the current, voltage or frequency values of power output beyond the operational values which can damage the equipment of consumers or cause a power outage.

A bulk majority of these faults arise from natural phenomena like lightning, erosion, high-speed wind, etc. as well as human activities which are usually not under control (Julio Barros & José M. Drake, 1994). Faults in an electrical power system are unavoidable making it a focal point of research in the field of power systems since the establishment of transmission and distribution systems (Qais Alsafasfeh et al., 2010). The importance of a well-coordinated system for the protection of the electrical grid against faults cannot be over-emphasized.

Faults in electrical power systems are generally classified into two broad categories. These are the symmetrical and unsymmetrical faults (Pavan Venkata et al., 2022).

1.1.1 Symmetrical Faults

Symmetrical faults occur in a power system when there is a short circuit connection across all the three conductors in a 3-phase line. This fault results in equal fault currents in the lines. Symmetrical faults occur as either a line-to-line-to-line (L-L-L) fault where all the three line conductors are brought into a short circuit connection without connection to the ground or a line-to-line-to-line-to-ground (L-L-L-G) fault where all the three line conductors including the ground are brought into a short circuit connection. L-L-L fault is depicted in figure 1.2 below.

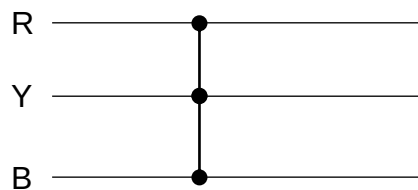


Figure 1.2: Line-to-Line-to-Line (L-L-L) Fault

L-L-L-G fault is depicted in figure 1.3 below.

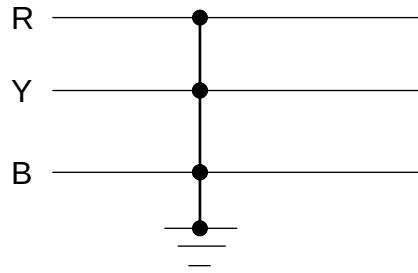


Figure 1.3: Line-to-Line-to-Line-Ground (L-L-L-G) Fault

1.1.2 Unsymmetrical Faults

Unsymmetrical faults in power systems occur when there is an unsymmetrical current across the 3 lines of conductors due to unequal magnitudes of faults along the lines. The majority of faults that occur in power systems are unsymmetrical faults with the most common type being the occurrence of short circuit from one phase to the ground (Halil Alper Tokel et al., 2018). Unsymmetrical faults in power systems are further divided into three categories. These are L-G faults, L-L faults and L-L-G faults.

1.1.2.1 Single Line-to-Ground (L-G) Faults

This category of faults occurs when there is a short circuit from one of the 3 phases to the ground. Red-to-ground (R-G), yellow-to-ground (Y-G) and blue-to-ground (B-G) are three types of faults that fall into this category. These faults are depicted in figure 1.4 below.

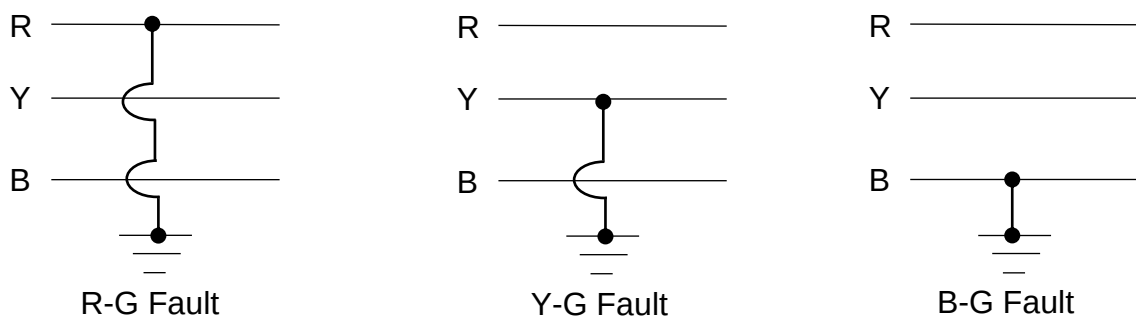


Figure 1.4: Single line-to-ground (L-G) faults

1.1.2.2 Line-to-Line (L-L) Faults

Line-to-line faults occur when there is a short circuit from one line to another line without affecting the third line or making a connection with the ground. The various types of faults that

fall into this category are red-to-yellow (R-Y) fault, yellow-to-blue (Y-B) fault and red-to-blue (R-B) fault. These faults are depicted in figure 1.5 below.

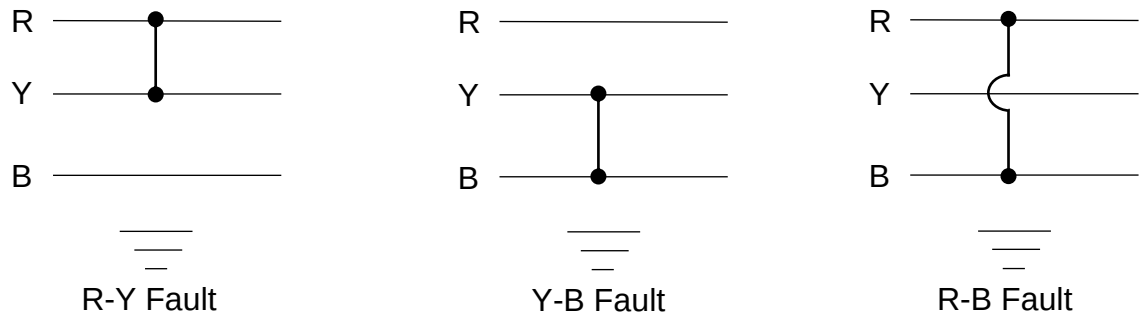


Figure 1.5: Line-to-Line (L-L) Faults

1.1.2.3 Line-to-Line-to-Ground (L-L-G) Faults

Line-to-line-to-ground faults are caused when there is a short circuit from one line to another line and also to the ground while the third line is not affected. The various types of faults that fall into this category are red-to-yellow-to-ground (R-Y-G) fault, yellow-to-blue-to-ground (Y-B-G) fault and red-to-blue-to-ground (R-B-G) fault. These faults are depicted in figure 1.6 below.

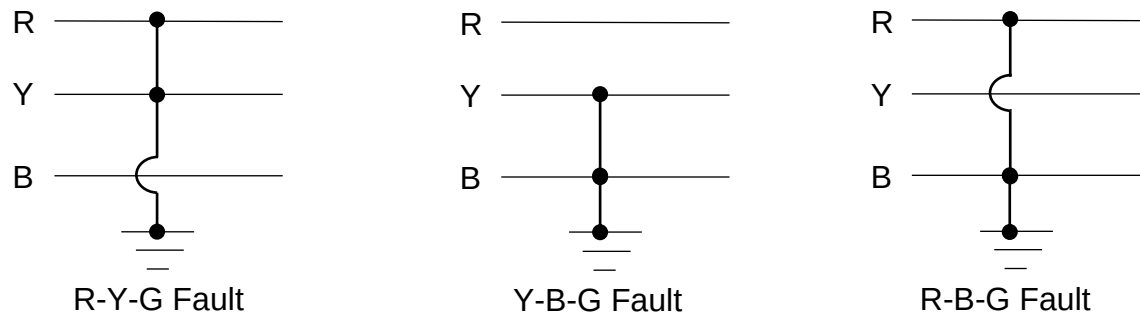


Figure 1.6: Line-to-Line-to-Ground (L-L-G) Faults

The effects of various faults on voltage and current as presented by (YUMURTACI et al., 2016) is shown in figure 1.7 below.

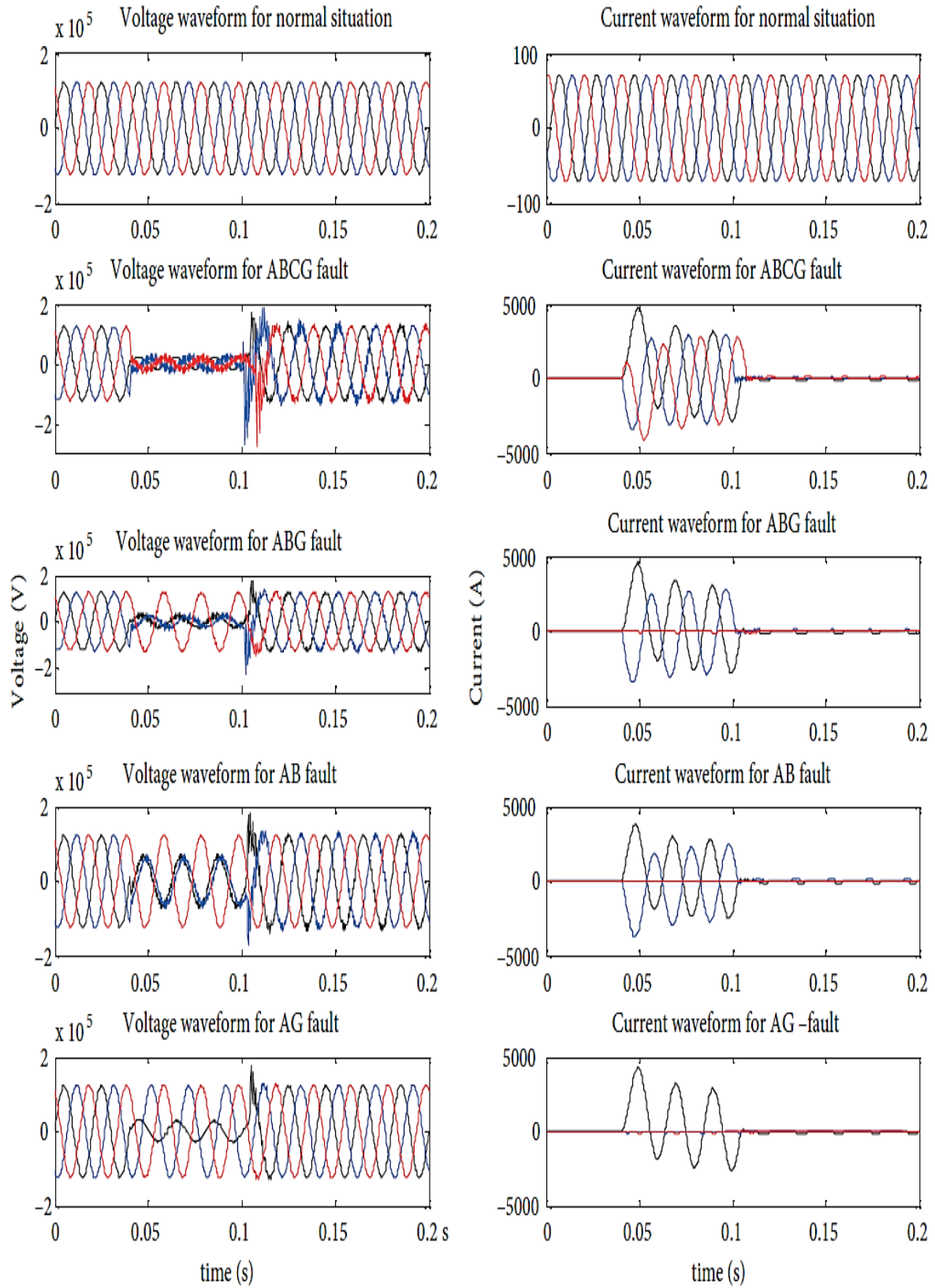


Figure 1.7: Effects of Various Faults on Voltage and Current. Source: (YUMURTACI et al., 2016)

1.2 Problem Statement

The reliability and continuous delivery of electricity to consumers are vital for modern society and power systems play a major role in ensuring these requirements are met (Mohammed Wadi, 2021). However, power systems are vulnerable to several factors which inhibit the safe delivery of electricity with faults being the major ones. These faults not only cause disruption to the continuous flow of electricity but can also damage equipment and compromise safety. Fault detection and classification play a vital role in the detection and identification of these faults. However, despite the various advancements in technology and contributions made by researchers, the ability to detect and classify faults as fast as possible is still a field which requires further research (Radu-Adrian Tirnovan & Maria Cristea, 2019). The following are some of the challenges being faced during the process of fault detection and classification.

1. **Diverse Fault types:** Faults experienced in power systems occur in a number of different ways which include but not limited to short circuit faults, open circuit faults, and transient faults (Qais Alsafasfeh et al., 2010). Each of these types of fault has unique characteristics and poses a unique type of challenge during fault detection and classification.
2. **Complex Network Architecture:** Power systems are comprised of several interconnected smaller networks of electrical components. This vast nature of power systems poses challenges in the form of fault propagation and the ability to isolate and recognize the source of faults (Radu-Adrian Tirnovan & Maria Cristea, 2019).
3. **Operation in a Fast-Changing Environment:** The power system is characterized by the fast and dynamic changes in network parameters like the pattern of power generation, variable load demand and network configurations. This dynamic nature of power systems requires fault detection and classification systems and algorithms to be very robust and adaptive to changes within the system (Khalfan Al Kharusi et al., 2022).
4. **Availability of Quality Data:** Fault detection and classification relies heavily on quality data from sensors and monitoring devices in order to function properly. However, the data from these sensors and monitoring devices sometimes have inconsistencies or missing values which make the fault detection and classification system give inaccurate results.
5. **Real-Time Response Requirement:** Fault are quick at causing damages to equipment in a power system. This requires the detection and implementation of protective measures in order to ensure the safety of the connected equipment. This requirement as vital as it

is, is still an area requiring further research (Radu-Adrian Tirnovan & Maria Cristea, 2019).

1.3 Aim of the Study

The aim of this project is to develop a procedure which requires minimal time for the detection of faults in the transmission system and also robust enough to work accurately in a dynamical and diverse range of electrical networks to quickly detect faults and implement protective measures ensuring the safety of connected electrical components.

1.3.1 Objectives of the Study

The objectives of this project are as follows:

1. To propose a new novel and scalable real-time technique for the detection of different fault scenarios.
2. To implement a technique for real-time identification of transmission line where fault has occurred.
3. To implement a technique for real-time identification of the type of fault that has occurred.
4. To implement a technique for getting the transmission line along which fault has occurred.
5. To implement a technique for getting the exact location of fault along a transmission line where fault has occurred.
6. To implement a relay system capable of disconnecting the loads (distribution systems) in real-time on the occurrence of faults to mitigate any adverse effects that could have been caused by the fault.

1.4 Justification of Study

Power System forms the backbone of any modernized society comprising of the generation, transmission and distribution of electrical power to meet the day-to-day demands of consumers. As vital as the power system is, it is still vulnerable to a number of environmental disturbances which are orchestrated in form of faults. These faults are most common in the transmission system according to (Soufiane Belagoune et al., 2021). This is because it is the most exposed to environmental disturbances such as lightning, accidents which are caused by human activities, corrosion, etc. According to (Julio Barros & José M. Drake, 1994) more than 80%

of all faults occurrence are caused by lightning. These faults are capable of interrupting the continuous flow of electricity, causing damages and endangering consumers. Fault detection and classification are essential tools to troubleshoot these fault occurrences in order to take protective actions and remedy the effects of these faults.

Fault detection and classification is a vital and core area in the operation and management of power systems (Qais Alsafasfeh et al., 2010). Quick detection and classification of faults is a requirement for quick repairs which ensure the reliability, robustness and continuity of electrical flow in power system.

This project aims to build on the various work done in the field of fault detection and classification in power system through the proposition of an alternative approach to quickly detect fault and take protective and remedial actions in real-time. This would benefit industry stakeholder and improve the reliability, robustness and sustainability of the electrical grid.

1.5 Significance of the Study

Despite the numerous works being done in the field of fault detection and classification in power system, this area is still faced with a lot challenges which includes but not limited to diverse fault types, different network configurations and dynamic operating conditions. This project aims to develop a technique capable of adapting to the dynamic and fast changing nature of electrical grids.

1.6 Scope and Limitation

The following are the scope and limitations of this project:

1. This project focuses on developing a novel approach to the fault detection phase only. The classification phase would follow similar ones to those found in previous researches.
2. This project is focused on the transmission system only. Any other part of the electrical grid is out of the scope of this project. Though same principles can be applied in the distribution system.
3. The proposed approach would require that no fault occurred during the first full cycle that is the first 16.6667 milliseconds. Some of the parameters used to detect faults would be learned at the completion of a full cycle. When fault occurs within the first

full cycle, wrong parameters would be learned and the system would not behave as expected.

4. This approach assumes that the voltage along the transmission line remains approximately constant. This approach is not applicable when the transmission voltage is meant to vary significantly during transmission.

CHAPTER TWO: REVIEW OF PAST WORK

2.1 Literature Review

Fault detection and classification is undoubtedly an indispensable part of power system analysis and has been a focal point of research in power systems since the emergence of electricity transmission and distribution systems (Qais Alsafasfeh et al., 2010). It would be of utmost importance to have an overview of the numerous work and contribution by researchers in the field of fault detection and classification in power system in order to understand how this field has evolved over time and the current state of the field.

Some of the earliest researches done in the field of fault detection and classification focuses on using classical methods to detect faults based on the waveforms of the line voltages and currents as proposed by (Julio Barros & José M. Drake, 1994). This approach consists of multiple stages. The first stage is to detect an overcurrent in any of the three-phase signals by utilizing the relative difference between the current of a cycle with that of the previous cycle. When an overcurrent is detected, the other stages would be activated. The next stage involves processing the voltage samples by a set of Kalman filters. This involves using two filters where one is initialized with the parameters which correspond to the assumed faulty phase while the other filter is initialized with the parameters which correspond to the nonfaulty phase. These filters are used to get the voltage phasors of the three phases based on the conditions assumed earlier. Finally, the probability of the occurrence of a fault is gotten using the voltage phasors and the distribution of possible values.

Another research done by (Fahmida N. Chowdhury & Jorge L. Aravena, 1998) during the earliest years gave rise to the proposition of a modular approach for fault detection and classification in power systems. This approach consists of three stages called Modules as shown in figure 2.1 below.

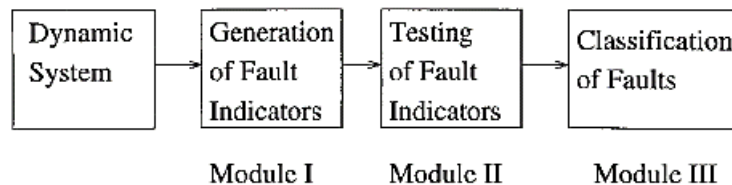


Figure 2.1: Fahmida and Jorge's Approach. Source: (Fahmida N. Chowdhury & Jorge L. Aravena, 1998)

Module I consist of two sub-approaches in generating the fault indicators; the model-based approach and the model-free approach. The model-based sub-approach is used when there is a model that can predict the nonfaulty output of the power system. The model can be a mathematical model that accurately gives the output of the nonfaulty system or an input-output (I/O) model of the system. In the presence of such a model, a residual would be calculated which is the difference between the output of the power system and the model as shown in figure 2.1 below.

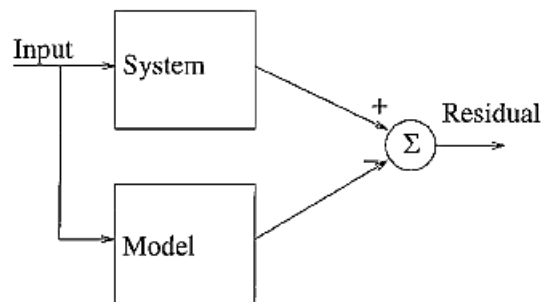


Figure 2.2: Module I of Fahmida and Jorge's Approach. Source: (Fahmida N. Chowdhury & Jorge L. Aravena, 1998)

The model-free sub-approach involves using multirate filter banks for the creation of an orthogonal decomposition. The multirate filter banks process the input signals then generate a set of fault indicators. After the fault indicators have been gotten, they would be fed as input into a three-layered neural network which predicts the occurrence of a fault.

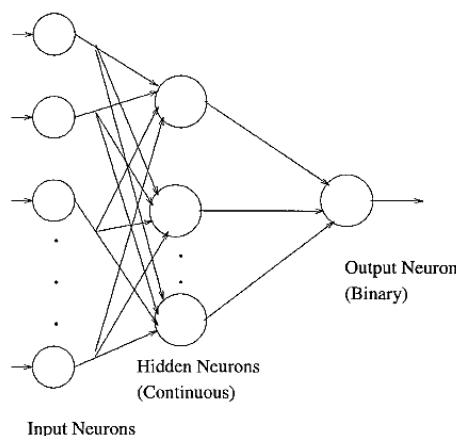


Figure 2.3: Neural Network Architecture in Fahmida and Jorge's Approach. Source: (Fahmida N. Chowdhury & Jorge L. Aravena, 1998)

If a fault is detected in Module II, the fault indicators from module I are fed as inputs into a self-organizing neural network (Kohonen Net) for the purpose of fault classification.

From the papers above, it can be seen that the research done in these eras either used a pure classical approach or a combination of feature extraction, fault prediction and fault classification. It is worthy of note that the Artificial Neural Networks used during this time are shallow networks. This is because the emergence of complex and deep neural networks has not begun yet.

Moving on to the last decade (2010 – 2019), a lot of work was also done in the field of fault detection and classification in power systems.

(Qais Alsafasfeh et al., 2010) proposes an electrical protective relaying framework that makes use of the first one-fourth cycle of phase current to detect and classify faults in an electrical power system. The approach involves first getting the difference signal ΔS which is the difference between the current signal and its previous reading at each one-fourth of the signal cycle. This difference signal would then be modelled as shown below:

$$A\Delta S + Bt + C = 0$$

A, B and C are constants which are derived from line specific intersect point. This equation would then be used to transform each of the values of the difference signal into magnitude and phase in the phasor domain. This would be followed by transforming the result into symmetrical components using the symmetrical component technique. The last step would result in a series for each of the positive and negative signatures computed over a time series within one-fourth of a cycle. These positive and negative signatures would be used as input to Principal Component Analysis (PCA) technique which project the inputs onto the principal component space. Thereafter, a measure of distance would be used for a similarity check. All the readings for this approach are taken at the sending end which eliminates the need for communication between the sending end and the receiving end.

(Aida Asadi Majd et al., 2017) proposed a k-NN-based approach for fault detection and classification. The fault detection process involves getting the squared normalized currents for all current values within a moving window of half-cycle length. Then k-NN algorithm would be used to get the fifth nearest neighbor to each value followed by calculating the distance which is the difference between each value and the fifth nearest neighbor. The maximum

distance would then be selected and compared to a predetermined threshold for fault detection. The process is illustrated in figure 2.4 below.

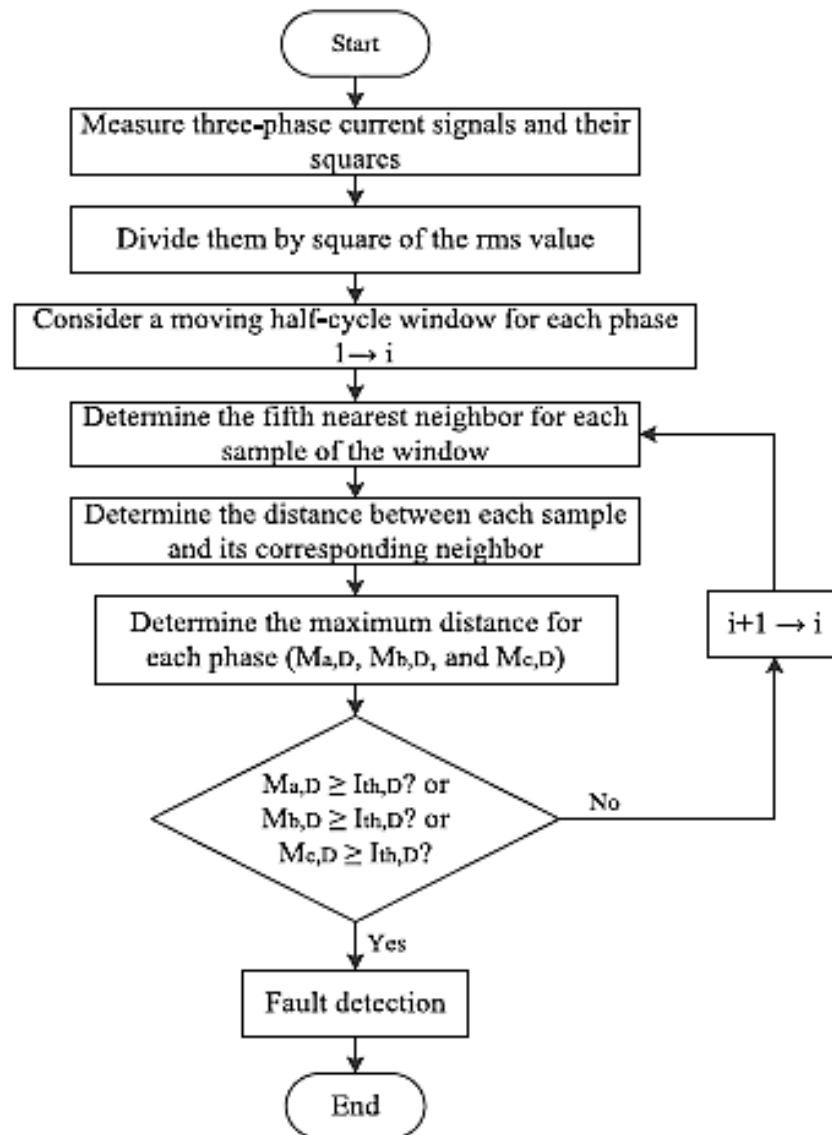


Figure 2.4: Flowchart of Approach Proposed by Aida Asadi Majd and his co-authors.

Source: (Aida Asadi Majd et al., 2017)

The fault classification process follows the same approach as fault detection however the instantaneous values of the three-phase currents after being normalized are used. The sliding window would also have a length of three-quarters of a cycle as opposed to half for fault detection.

A typical architecture of a Wide Area Measurement System (WAMS) is shown in figure 2.5 below.

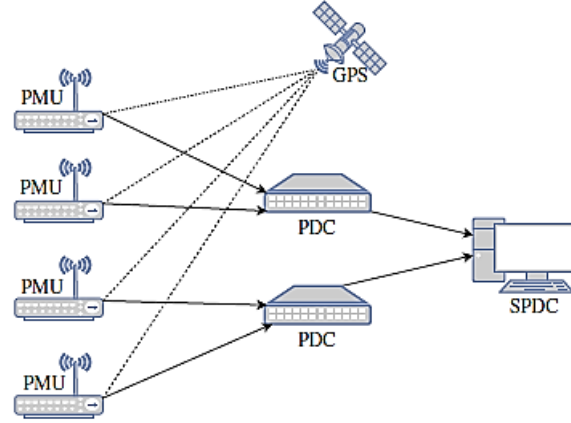


Figure 2.5: Architecture of a Wide Area Measurement System (WAMS). Source: (Halil Alper Tokel et al., 2018)

In WAMS, the Phasor Data Concentrators (PDCs) align incoming signals by the timestamps ensuring synchronization before being sent to the Super Phasor Data Concentrator (SPDC). However, in some scenario signals arrive later than the wait time which the PDCs wait for signals which affects the synchronization of data as shown in figure 2.6 below.

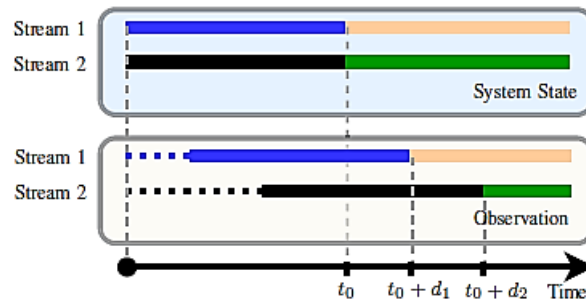


Figure 2.6: Delay in Data Arrival. Source: (Halil Alper Tokel et al., 2018)

This prompted (Halil Alper Tokel et al., 2018) to propose a methodology which is focused on the effect of the communication network delays on fault detection and classification while taking delayed measurements into consideration.

(Mohammed Musa et al., 2019) proposes an approach of using the covariance indices for fault detection and classification where it was found that the covariance indices for the faulty phase are much larger than zero while the covariance indices of the healthy phase are zero.

An approach which utilizes the Stockwell Transform was proposed by (Gustavo Santos et al., 2019). The approach involves periodically sampling the three-phase current signals and

transforming the signal into the Stockwell domain. Thereafter, the transformed signals are analyzed to detect fault. If a fault is detected, the current and voltage data would be processed and passed through a multilayer perceptron classifier to classify the fault. The approach is illustrated in figure 2.7 below.

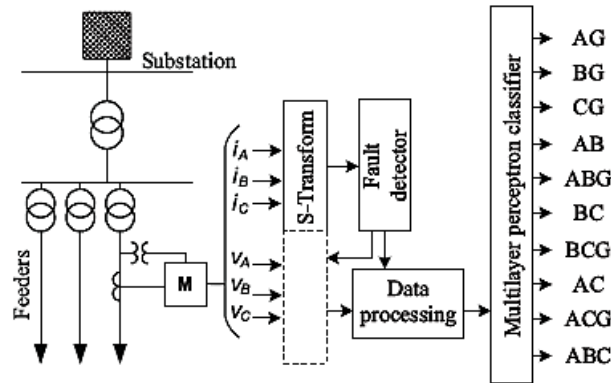


Figure 2.7: Visualization of the Approach of Gustavo Santos and his co-authors. Source: (Gustavo Santos et al., 2019)

(Radu-Adrian Tirnovan & Maria Cristea, 2019) provided an overview of the techniques used for fault detection and classification in electrical power transmission systems. It was shown that the processes of fault detection, fault classification and fault location are usually in the following sequence of steps:

1. Current and Voltage Signal Acquisition
2. Feature Extraction
3. Fault Detection
4. Fault Classification
5. Fault Location

The processes are depicted in figure 2.8 below.

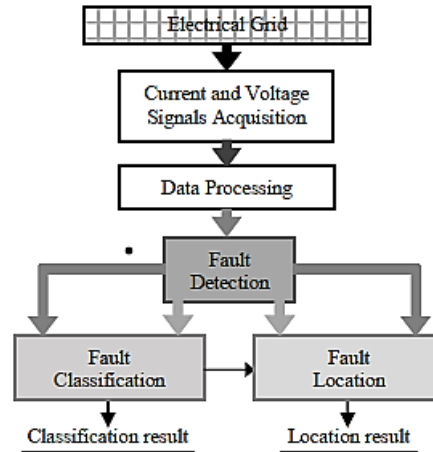


Figure 2.8: Processes of Fault Detection, Classification and Location. Source: (Radu-Adrian Tirnovan & Maria Cristea, 2019)

It was also shown that the various types of feature extraction techniques are done in:

1. Time Domain: such as using the RMS, mean, variance, crest factor, peak-to-peak, auto correlation, cross correlation, etc.
2. Frequency Domain: such as using spectral analysis, envelope analysis, spectrum analysis and higher order spectrum.
3. Time-Frequency Domain: such as using Short-Time Fourier Transform - STFT, localized Fourier transform (Gabor transform), S transform - ST, Empirical mode decomposition – EMD, Spectral kurtosis, etc.
4. Wavelet Domain: such as using Continuous wavelet transform (CWT), Discrete wavelet transform (DWT), Wavelet packet transform and Hilbert-Huang transform.
5. Data Descriptive Model: such as using Regression: Curve fitting, Autoregressive (AR) models, etc.
6. Time-Independent Transforms: Principal component analysis - PCA, Independent component analysis – ICA, etc.

The various approaches to fault classification in electrical power transmission systems using Artificial Intelligence (AI) were also categorized into two which are:

1. Pattern Recognition: This set of approaches involves combining the symmetrical component technique with the Principal Component Analysis (PCA) as was done by (Qais Alsafasfeh et al., 2010) or using the Multiresolution Analysis (MRA).

2. Machine Learning: This set of approaches involves using Decision Trees (DT), Artificial Neural Network (ANN), Support Vector Machine (SVM), Bayesian Networks, k-Nearest Neighbours (k-NN), Deep Learning, etc.

Despite the various research that has been done in the field of fault detection and classification in power systems, the goal of detecting faults as fast as possible while maintaining high robustness and accuracy still requires further research (Radu-Adrian Tirnovan & Maria Cristea, 2019). There are still a lot of research being done in the current decade (2020 till date) in the field of fault detection and classification in power system.

The sign of the change of the positive sequence current at the two ends of transmission lines was utilized by (Sauvik Biswas & Paresh Kumar Nayak, 2020) to detect fault while using the transient monitor index gotten from the three-phase current measured locally for fault classification.

(Abdulaziz Aljohani, 2020) proposed an approach of using Artificial Neural network (ANN) and Stockwell Transform (ST) for the detection and classification of single line to ground faults in a simulated motor power distribution center. The approach makes use of sinusoidal current signals which were obtained from a simulated motor power distribution center. Stockwell Transform would then be used on the signals to extract two matrices S and E. From these two matrices, four vectors would be calculated. These are:

1. Phase Contour Column (Pcc)
2. Maximum Column Values of the S Matrix (Scmax)
3. Maximum row Values of S Matrix (Srmax)
4. Maximum Column Values of E Matrix (Ecmax)

After the vectors above are gotten, 28 features would be extracted. These features are described in table 2.1 below.

Feature Number	Description	Feature Number	Description
1	Standard Deviation of Scmax	2	Maximum element of Pcc
3	Standard Deviation of Srmax	4	Maximum element of Ecmax
5	Standard Deviation of Pcc	6	Minimum element of Scmax
7	Standard Deviation of Ecmax	8	Minimum element of Srmax
9	Maximum element of Scmax	10	Minimum element of Pcc
11	Maximum element of Srmax	12	Minimum element of Ecmax
13	Mean of Scmax	14	Mean of Srmax
15	Mean of Pcc	16	Entropy of Srmax
17	Mean of Ecmax	18	Entropy of Scmax
19	Entropy of Pcc	20	Entropy of Ecmax
21	Skewness of Scmax	22	Skewness of Srmax
23	Skewness of Pcc	24	Skewness of Ecmax
25	Kurtosis of Scmax	26	Kurtosis of Srmax
27	Kurtosis of Pcc	28	Kurtosis of Ecmax

Table 2.1: Features Used in Approach Presented by Abdulaziz Aljohani. Source: (Abdulaziz Aljohani, 2020)

These features would then be used as inputs to a Multilayer Perceptron Artificial Neural Network (MLP-ANN).

The detection of phase to ground faults under high resistance ground paths remains a challenging task for protection engineers since the resulting current remains below the detection threshold of conventional protection relays. (Nakho et al., 2021) proposes a combination of Discrete Wavelet Transform (DWT) and k-Nearest Neighbor (k-NN) machine learning algorithm to detect and classify faults under high resistance ground path as a solution to this problem. There are three processes per phase in this approach as shown in figure 2.9 below.

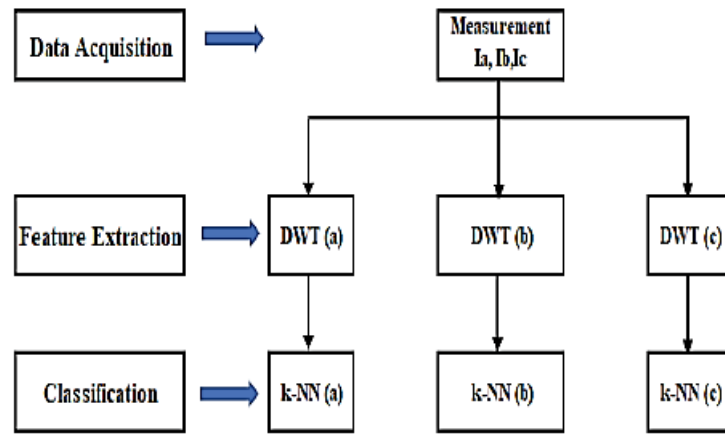


Figure 2.9: Three Processes Involved in the Approach of Nakho and his co-authors. Source: (Nakho et al., 2021)

(Mohammed Wadi, 2021) proposes two machine learning models each for two approaches to improve the process of fault detection in smart grids based on the fault detection data made available by the Technical University of Ostrava. The first proposed model is based on the concept of balancing imbalanced datasets into a balanced data set. The second proposed model is based on balancing imbalanced datasets into balanced datasets then applying the hyperparameter tuning approaches after the optimal features have been chosen. This research shows that a balanced dataset can achieve up to 96% better results than an imbalanced dataset.

(Soufiane Belagoune et al., 2021) proposed three deep learning classification and regression models based on deep recurrent neural networks for fault region identification, fault type classification and fault location prediction. Sequential Deep Learning (SDL) was employed through Long Short-Term Memory (LSTM). The algorithms were thereafter tested in a two-area four-machine power system.

(Ragab Mahmoud, 2022) presented a digital protection scheme based on three indices derived from the three-phase input and output current signals of the protected equipment, to detect current disturbances, classify faults, discriminate fault directions, select faulty phase(s), define current unbalances, and identify current transformer saturation extent. Moreover, it develops a new closed-tripping characteristic with a quadrilateral form for differentiating the fault location (internal/external) with respect to the element protection zone. The flow chart for this approach is shown in figures 2.10 and 2.11 below.

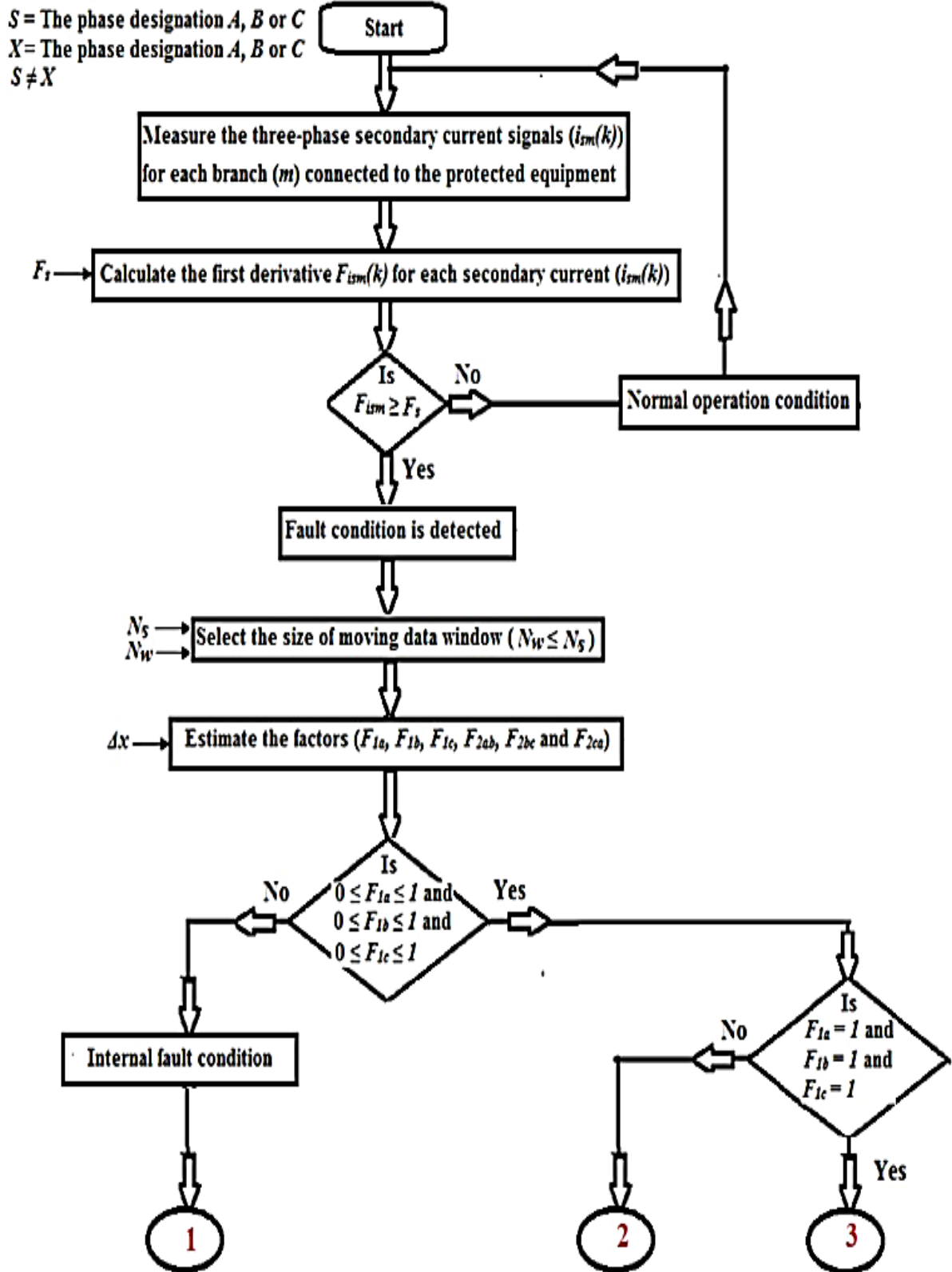


Figure 2.10: Flowchart for the Approach of Ragab Mahmoud (Part I). Source: (Ragab Mahmoud, 2022)

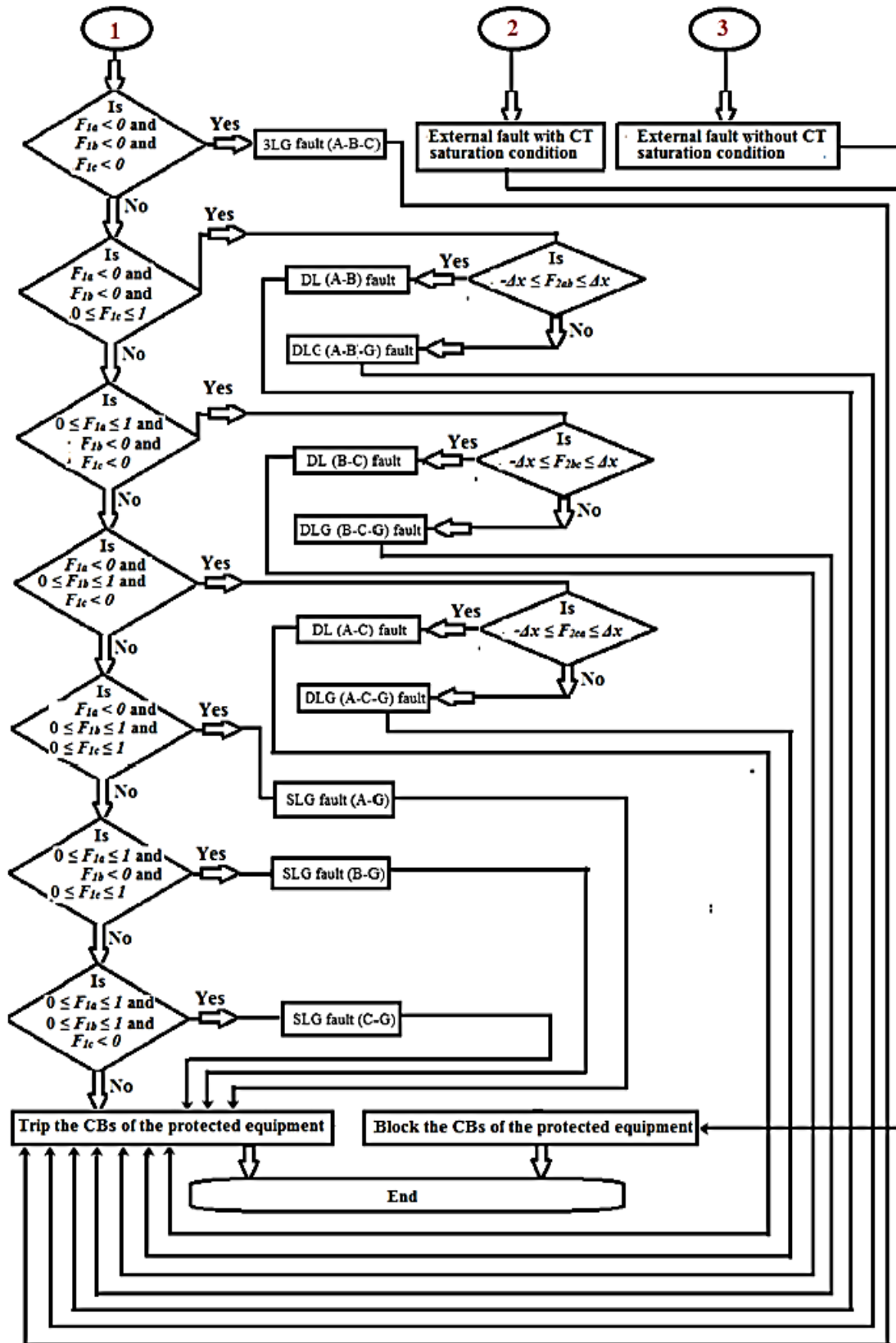


Figure 2.11: Flowchart for the Approach of Ragab Mahmoud (Part II). Source: (Ragab Mahmoud, 2022)

(Khalfan Al Kharusi et al., 2022) proposes a machine learning approach to detect and classify faults in transmission lines connected to inverter-based generators. This approach makes use of a two-layer classification scheme where the first layer detects faults and upon the detection of faults by the first layer, the second layer is activated to classify the fault. The features from measured three-phase voltages and currents were extracted in the time, frequency, and time-frequency domains.

(Pavan Venkata et al., 2022) proposed a support vector machine (SVM) model for the detection and classification of faults in a transmission system utilizing the time and frequency series parameters at both ends of a line computed over the most recent eight samples. The model makes use of the quarter-cycle data of power frequency to detect and classify faults. The computed time and frequency series parameters are fed into a Support Vector Machine (SVM) model which detects and classifies faults. The processes involved in this approach are illustrated in figure 2.12 below.

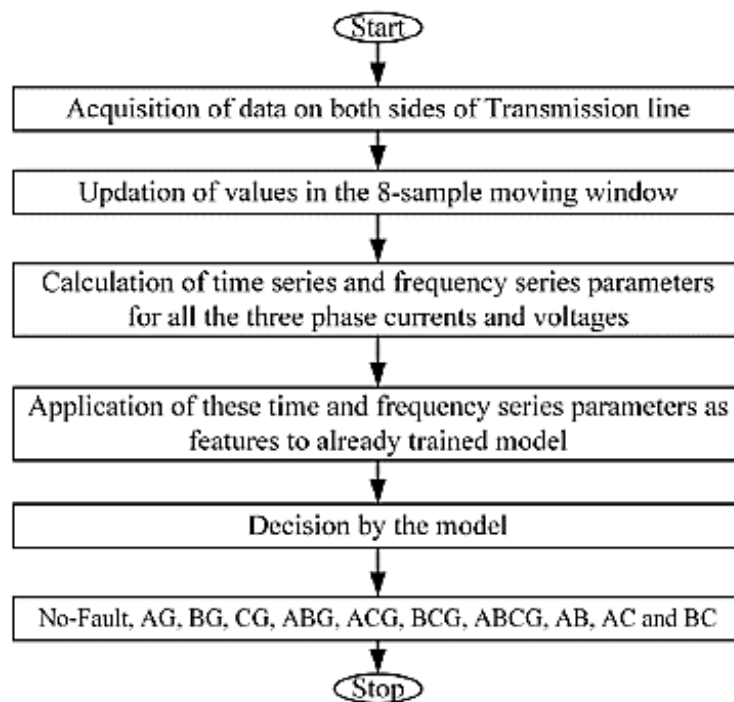


Figure 2.12: Processes Involved in the Approach of Pavan Venkata and his co-authors.

Source: (Pavan Venkata et al., 2022)

(Zahra Yahyaoui et al., 2022) proposed a Bidirectional Long Short-Term Memory (BiLSTM) approach based on Kernel Principal Component Analysis KPCA for wind energy conversion system fault detection and diagnosis. The approach extracts and selects from the features using

KPCA after which the selected features are fed into the BiLSTM model for classification. The approach was validated using data gotten from healthy wind energy conversion converters that were injected with several fault scenarios.

(Ehab M. Esmail et al., 2023) proposes two schemes for simultaneous series and shunt earth fault detection and classification involving two stages. The former being the fault detection and identification stage while the latter being the fault classification stage. The first scheme involves communication links between both ends. The second scheme involves fault classification units at both ends which applies the Clarke transform on local current signals to classify the open conductor and simultaneous faults. The processes involved in this approach are shown in figure 2.13 below.

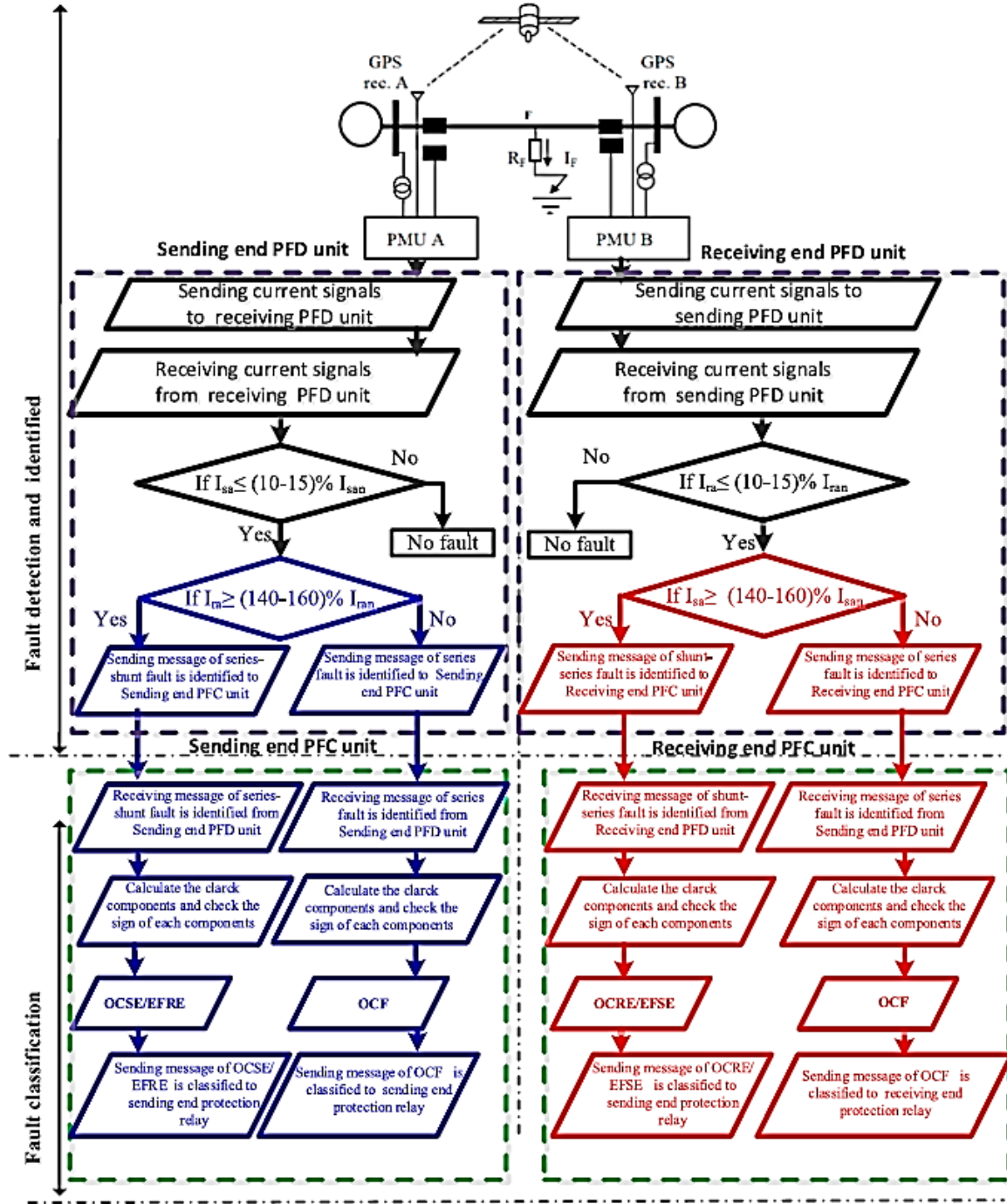


Figure 2.13: Processes Involved in the Approach of Ehab M. Esmail et al. Source: (Ehab M. Esmail et al., 2023)

(B. Suresh Kumar et al., 2023) demonstrated the effectiveness of using MATLAB and Machine learning to simulate and detect transmission line faults as well as classifying them with high accuracy.

(Juan Pablo Naranjo Cuéllar et al., 2023) proposed a technique that uses the geometry of the curve generated by the Clarke transform of the three-phase voltages of power systems to classify different types of short circuit faults for real-time detection. It was shown that the eigenvalue of the matrix that characterizes an ellipse associated with each fault can be used to determine the fault inception distance while the corresponding eigenvector can be used to identify the type of fault. A machine learning model which was embedded into a Raspberry Pi 3 was built based on this characterization method.

(Nandhini & Prajith, 2023) did a review of the latest machine learning methods used for the identification and classification of faults in power systems. It was shown that varying the parameters according to power system topology is needed while using the wavelet transformation for fault classification in transmission lines. This also requires choosing the mother wavelet analysis when analyzing fault. The mother wavelet is mostly chosen to be the Daubechies or Morlet wavelet. This led to the need for a more general fault classification. Consequently, machine learning methods were chosen due to their generalization property. The data for training the machine learning models can be extracted using any of the signal processing methods such as wavelet transformation, fast Fourier transform or Hilbert Huang Transform, etc. Some methods use the energy values from the faulty signals to convert them into a proportional pixel value, where these image matrices are used as features for fault analysis.

2.2 Summary of Literatures

Author(s)	Title	Year	Methodology
Julio Barros and José M. Drake	Realtime fault detection and classification in power systems using microprocessors	1994	Estimation of the three-phase voltage phasors by means of a set of Kalman filters, and on the calculation of the fault probability
Fahmida N. Chowdhury	A modular methodology for fast fault detection and	1998	Use of multirate filter banks based on wavelet decomposition of actual data

and Jorge L. Aravena	classification in power systems		
Qais Alsafasfeh, Ikhlas Abdel-Qader and Ahmad Harb	Symmetrical pattern and PCA-based framework for fault detection and classification in power systems	2010	Use an integrated method that combines symmetrical components technique with principal component analysis
Aida Asadi Majd, Haidar Samet and Teymoor Ghanbari	K-NN-based fault detection and classification methods for power transmission systems	2014	Use of k-NN algorithm to get the distance between signal samples and the fifth nearest neighbor
Halil Alper Tokel, Rana Al Halaseh, Gholamreza Alirezai and Rudolf Mathar	A new approach for machine learning-based fault detection and classification in power systems	2018	Focus on the effect of the communication network delays on fault detection and classification which takes delayed measurements into consideration
Mohammed Musa, Zhengyou He, Ling Fu and Yujia Deng	A covariance-indices-based method for fault detection and classification in a power transmission system during power swing	2019	Use of covariance indices for fault detection and classification
Gustavo G. Santos, Thiago Menezes, Jose Vieira and Pedro Barra	An S-Transform-based approach for fault detection and classification in power distribution systems	2019	Use of Stockwell Transform and the energy concept to extract a group of even harmonics, from 12 to 42 as well as the fundamental frequency of the signals for the detection of fault
Radu-Adrian Tirnovan and Maria Cristea	Advanced techniques for fault detection and classification in electrical	2019	Overview of the methods used for fault detection and classification in

	power transmission systems: An overview		transmission lines and distributed systems
Sauvik Biswas and Paresh Kumar Nayak	A Fault Detection and Classification Scheme for Unified Power Flow Controller Compensated Transmission Lines Connecting Wind Farms	2020	Utilizing the sign of the change in the magnitude of positive-sequence current at both ends of the transmission line for fault detection and the transient monitor index derived from the locally measured three-phase current for fault classification
Abdulaziz Aljohani	Centralized Fault Detection and Classification for Motor Power Distribution Centers Utilizing MLP-NN and Stockwell Transform	2020	Use of Artificial Neural Networks (ANN) and Stockwell Transform to detect and classify Single line-to-ground (SLG) faults in a simulated motor power distribution center
A. Nakho, Yskandar Hamam and Y. Hamam	Detection and Classification of Single Phase to Ground Faults Under High Resistance Ground Paths in Power Systems using Machine Learning	2021	A combination of discrete wavelet transform and k-Nearest Neighbor machine learning algorithm to detect and classify faults under high resistance ground path as a solution to this problem
Mohammed Wadi	Fault detection in power grids based on improved supervised machine learning binary classification	2021	Machine learning models for two approaches to improve the process of fault detection in smart grids based on the fault detection data made available by the Technical University of Ostrava
Soufiane Belagoune, Nouredine	Deep learning through LSTM classification and regression for	2021	Use of three deep learning classification and regression models based on deep recurrent neural

Bali, Azzeddine Bakdi, Bousaadia Baadji and Karim Atif	transmission line fault detection, diagnosis and location in large-scale multi-machine power systems		networks for fault region identification, fault type classification and fault location prediction
Ragab A. Mahmoud	Simple algorithm for fault detection, classification and direction discrimination in power systems based on digital current signals processing	2022	Digital protection scheme based on three indices derived from the three- phase input and output current signals of the protected equipment, to detect current disturbances, classify faults, discriminate fault directions, select faulty phase(s), define current unbalances, and identify current transformer saturation extent
Khalfan Al Kharusi, Abdelsalam El Haffar and Mostefa Mesbah	Fault Detection and Classification in Transmission Lines Connected to Inverter- Based Generators Using Machine Learning	2022	Extraction of features from measured three-phase voltages and currents were extracted in the time, frequency, and time- frequency domains used for machine learning model to classify faults in transmission lines connected to inverter-based generators
Pavan Venkata, Vivek Pandya, Kushal Vala and Amit V. Sant	Support vector machine for fast fault detection and classification in modern power systems using quarter-cycle data	2022	Use of support vector machine (SVM) model for the detection and classification of faults in a transmission system utilizing the time and frequency series parameters at both ends of a line computed over the most recent eight samples
Zahra Yahyaoui,	Effective Fault Detection and Diagnosis for Power	2022	Use of Bidirectional Long Short- Term Memory (BiLSTM) approach

Mansour Hajji, Majdi Mansouri, Kamaleldin Abodayeh, Kais Bouzrara and Hazem Nounou	Converters in Wind Turbine Systems Using KPCA-Based BiLSTM		based on Kernel Principal Component Analysis KPCA for wind energy conversion system fault detection and diagnosis
Ehab M. Esmail, Faisal Alsaif, Shady H. E. Abdel Aleem, Almoataz Y. Abdelaziz, Anamika Yadav and Adel El- Shahat	Simultaneous series and shunt earth fault detection and classification using the Clarke transform for power transmission systems under different fault scenarios	2023	Use of communication links between both ends for one scheme and use of fault classification units at both ends which applies the Clarke transform on local current signals to classify the open conductor and simultaneous faults for another scheme
B. Suresh Kumar, V. Abhinav Varma, S. Devika Rani and A. Nishanth	Fault Detection and Classification using Machine Learning	2023	Design of transmission line in MATLAB SIMULINK and the use of fault toolbox to create faults used to train machine learning models to get the best algorithm with the best result
Juan Pablo Naranjo Cuéllar, Gustavo Ramos López and Luis	Learning the geometry of short-circuit faults in power systems for real- time fault detection and classification	2023	Use of the geometry of the curve generated by Clarke transform of the three-phase voltages of power system to classify different types of short circuit faults for real time detection

Felipe Giraldo Trujillo			
K Nandhini and C A Prajith	Review on Fault Detection and Classification in Transmission Line using Machine Learning Methods	2023	Review of the latest machine learning methods used for the identification and classification of faults in power systems

Table 2.2: Summary of Literatures

2.3 Contribution to Research

In this project, a newer and faster approach for fault detection is presented. This new approach makes use of moving variances and moving averages of the voltage waveform to detect faults in the transmission system. With this new approach faults in transmission systems are detected faster than the approaches presented earlier in literatures.

CHAPTER THREE: METHODOLOGY

3.1 Intuition behind Fault Detection

When a fault occurs, the most pronounced characteristic of such a fault is the surge in current and reduction in voltage as explained by (Julio Barros & José M. Drake, 1994; Sayanee Das & Shuma Adhikari, 2020). The rate at which this surge in current and reduction in voltage occurs is a function of the type of fault that has occurred as well as the relative location where the fault has occurred. The currents of a nonfaulty and faulty power supply are shown in figure 3.1 below.

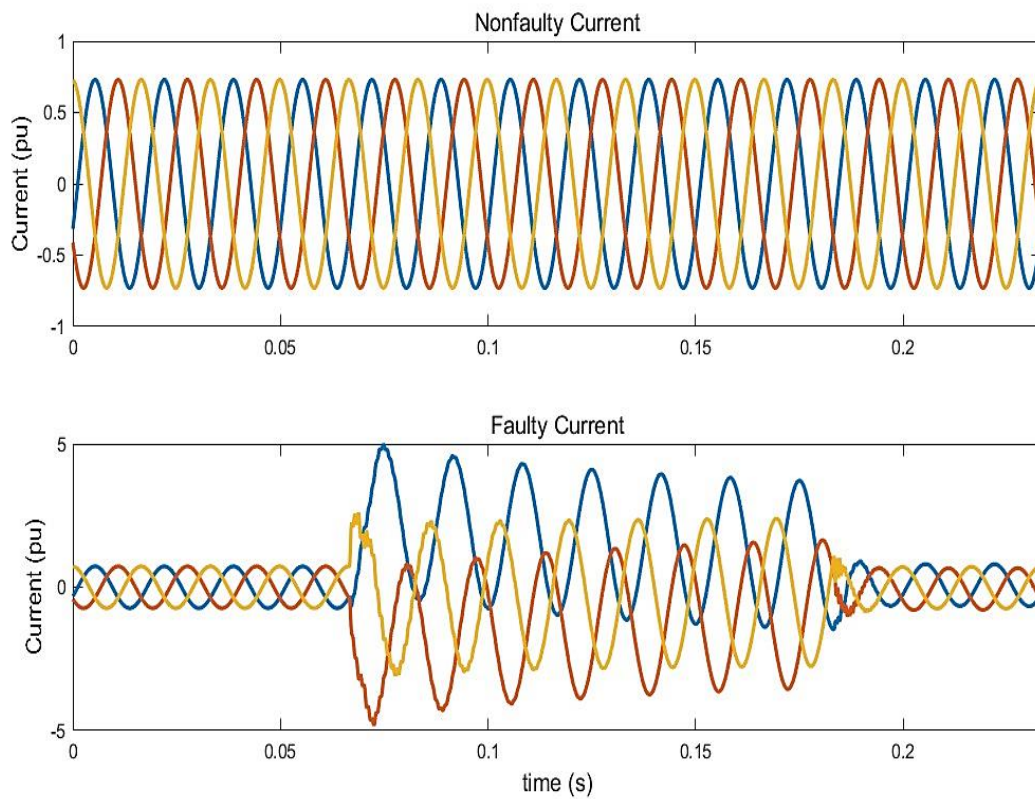


Figure 3.1: Effect of Fault on Current

The voltages of a nonfaulty and faulty power supply are shown in figure 3.2 below.

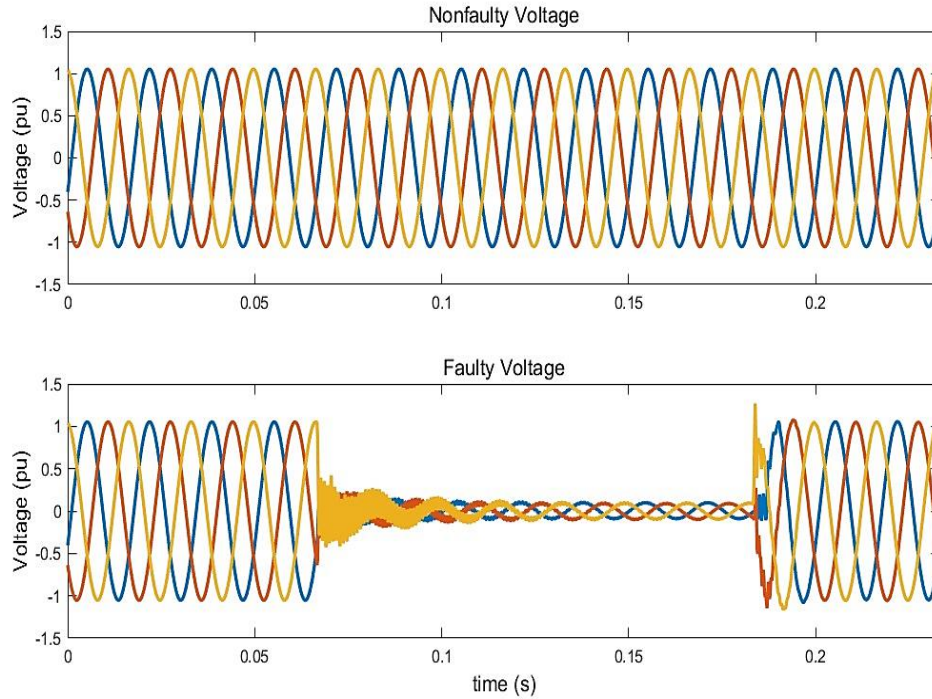


Figure 3.2: Effect of Fault on Voltage

3.2 Voltages as Parameters for Fault Detection

In a transmission system, the voltage is always kept at a constant value usually 230kV. This shows that the voltage would be preferred over the current as a parameter for fault detection. This is because the current in transmission lines unlike the voltages are not kept at constant value i.e., they decrease and increase as power increases and decreases.

3.3 Fault Indicator

The occurrence of a fault is always pronounced in the waveform of such a fault. A fault in reality has two major effects on the waveforms:

1. Increased or Reduced Amplitude of Waveform: A fault would cause the current waveform to have increased amplitude and on the other hand would cause the voltage waveform to have a reduced amplitude.
2. Shift in Central Position: It was observed during the course of carrying out this research that some faults give rise to shifts in the central position of waveforms.

In order to detect the occurrence of a fault using the effects of faults mentioned above, two indicators were adopted.

3.3.1 Moving Variance

The moving variance was adopted as an indicator to detect when the waveform has increased or decreased in amplitude. The moving variance was adopted over the moving standard deviation in order to amplify the changes in the spread of the waveform. The moving variance is calculated over a moving frame of a wavelength.

3.3.1.1 Fault Detection Using Moving Variance

In this proposed approach, the moving variance of each line at each bus would be stored when a full cycle is completed. For an electrical system with a frequency of 60 Hertz, a full cycle is completed at 16.6667 milliseconds after the system starts. When a voltage is detected having a moving variance of 0.03 pu lesser than the value when a full cycle is completed, such a waveform would be categorized as being faulty thereby triggering the fault occurrence indicator.

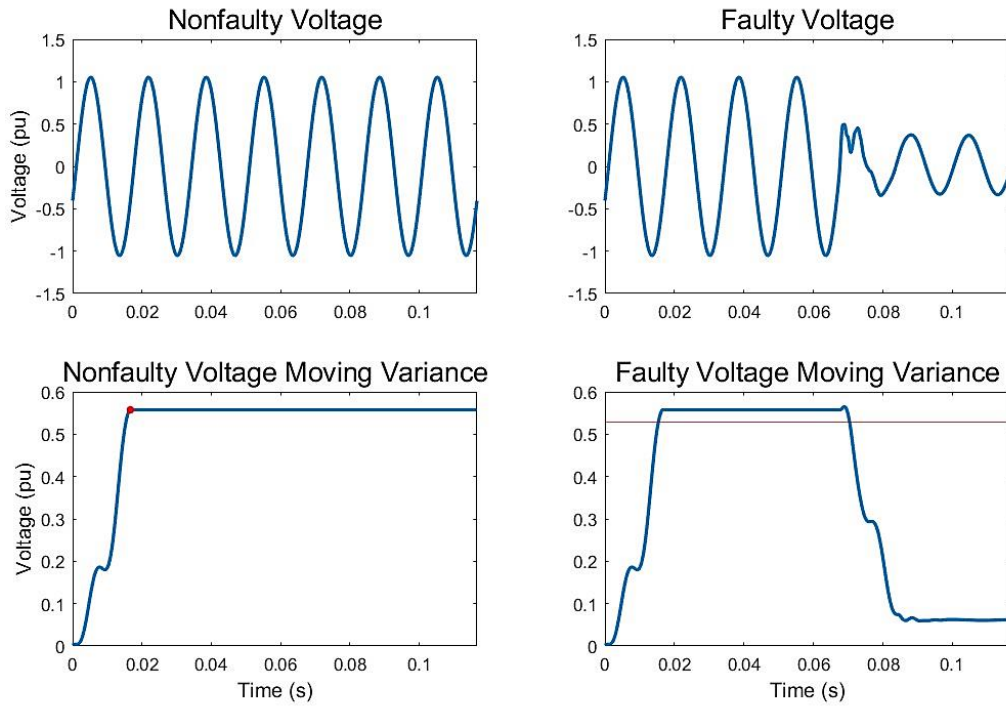


Figure 3.3: Fault Detection Using Moving Variance

3.3.2 Moving Average

The moving average was adopted as the second indicator to detect the changes in the central alignment of the waveform i.e. when the waveform moves up or down from the normal position. The moving average is also calculated over a moving frame of a wavelength.

3.3.2.1 Fault Detection Using Moving Average

In the proposed approach the moving average of each line at each bus would be calculated. Unlike the moving variance, this value would be used for comparison only and not stored when a full cycle is completed. Since the moving average is calculated over a wavelength the value should always be zero. Once a voltage with a moving average whose absolute value is more than 0.03 pu is detected, such a voltage would be classified as a faulty voltage triggering the fault detection indication.

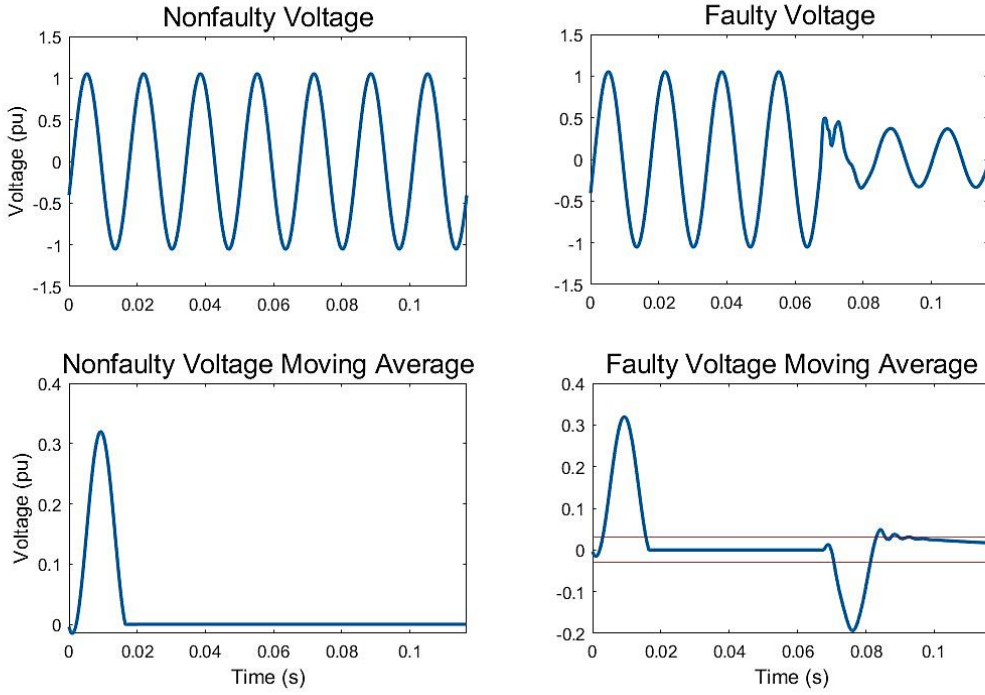


Figure 3.4: Fault Detection Using Moving Average

3.4 Data Generation

The procedure presented in this research work was tested on a MATLAB SIMULINK model of the IEEE 9 bus transmission system. The model used for this research was uploaded to MATLAB Central by (jaikumar, Pettikkattil, 2014) with over 20,000 downloads. The model consists of 9 buses as the name suggests. bus 1, bus 2, and bus 3 are the buses at the three generation stations in the model. The remaining buses are connected in a ring connection, forming the transmission lines in the middle. Furthermore, there are three loads in the model connected to buses 5, 6 and 8 respectively.

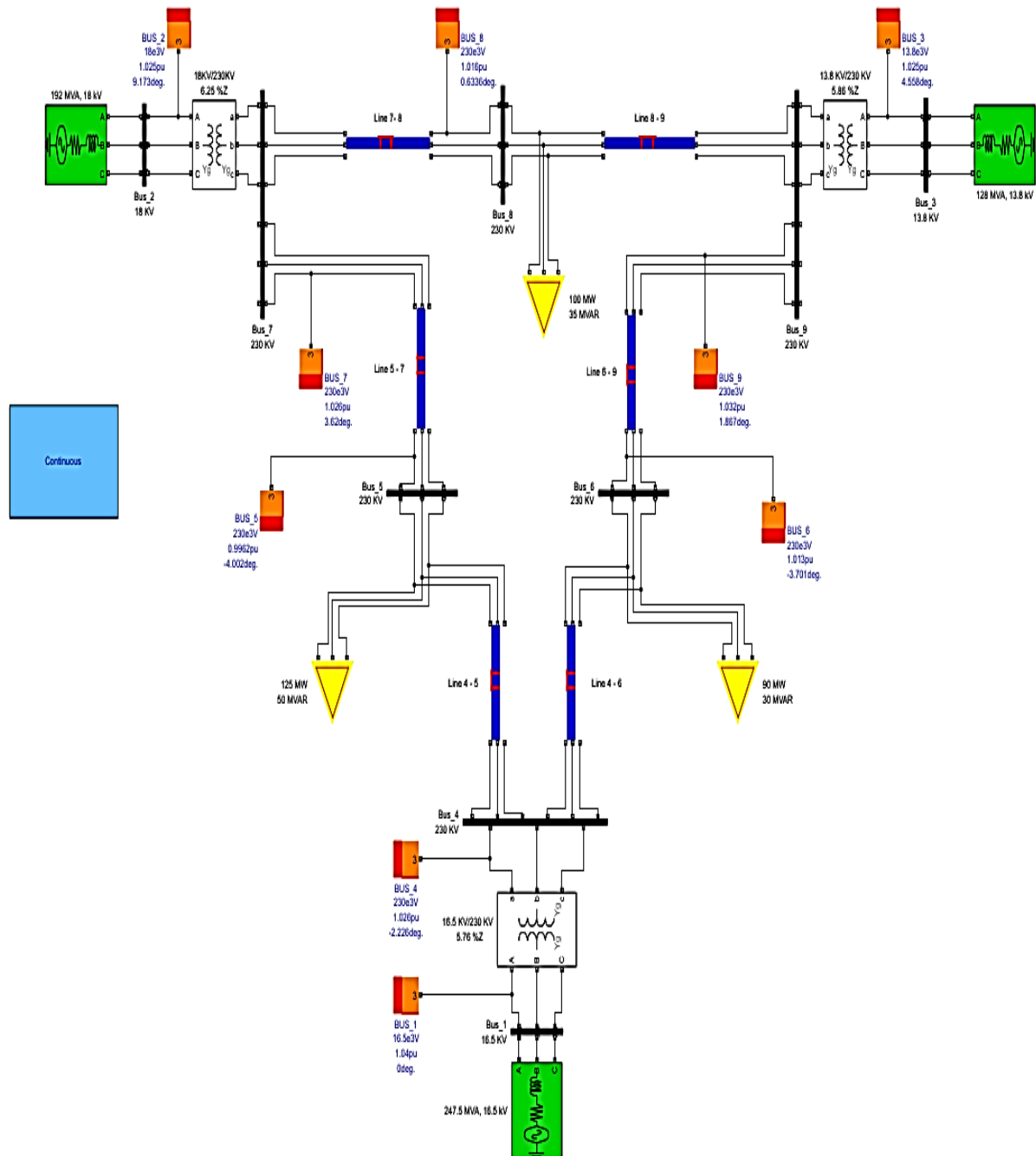


Figure 3.5: IEEE 9 Bus Transmission System Downloaded from MATLAB Central

3.4.1 Modified IEEE 9 Bus Transmission System

The downloaded model was not used as downloaded. Some modifications were made to the model in order to fit it to the fault detection and classification task.

3.4.1.1 Splitting of Three-Phase PI Section

The three-phase PI sections coloured as blue in the model in figure 3.5 above represent transmission lines each of which is 10km in length. These blocks were split into two parts to provide room for adding faults along the transmission lines. Each transmission line was split into two parts. The lengths of this new parts are varied in order to model faults at varying distances. However, the sum of the lengths of the two parts also equals 100km as the initial transmission line length

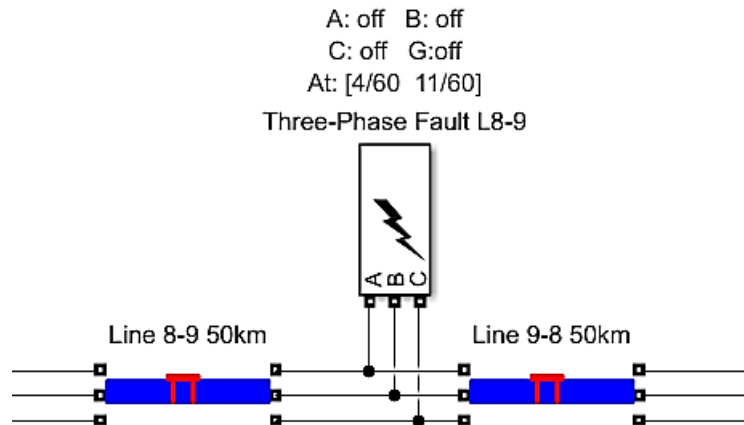


Figure 3.6: Splitting of the PI Sections (Transmission Lines).

3.4.1.2 Addition of Three-Phase Faults

After splitting the PI section representing the transmission lines, three-phase fault blocks were added between the transmission lines as shown in figure 3.6 above.

3.4.1.3 Addition of Three-Phase Circuit Breaker

Three-phase circuit breaker is added to each of the 3 loads in the transmission system in order to safeguard the consumers from the effects of the faults in the transmission lines.

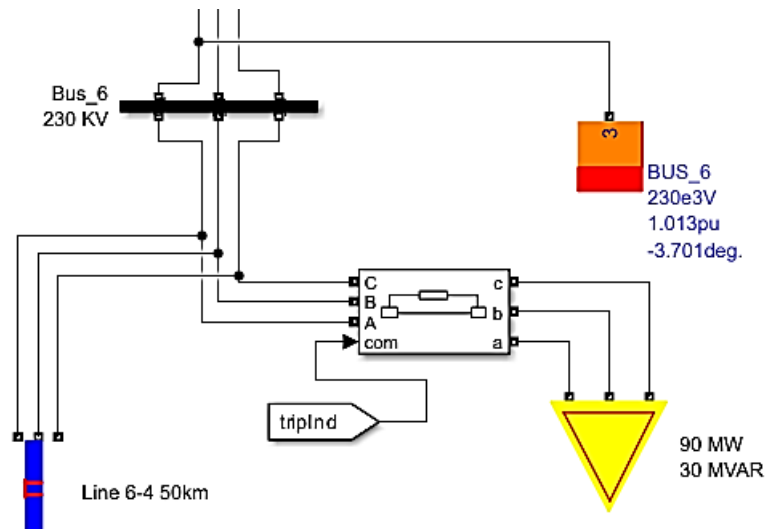


Figure 3.7: Addition of Three-Phase Circuit Breaker

3.4.1.4 Addition of Subsystems

Additionally, subsystems were added each for the processes of fault detection, data collection, fault line identification, fault type identification, fault location and relay system.

The modified IEEE 9 bus Transmission SIMULINK model used for this research is shown in figure 3.8 below.

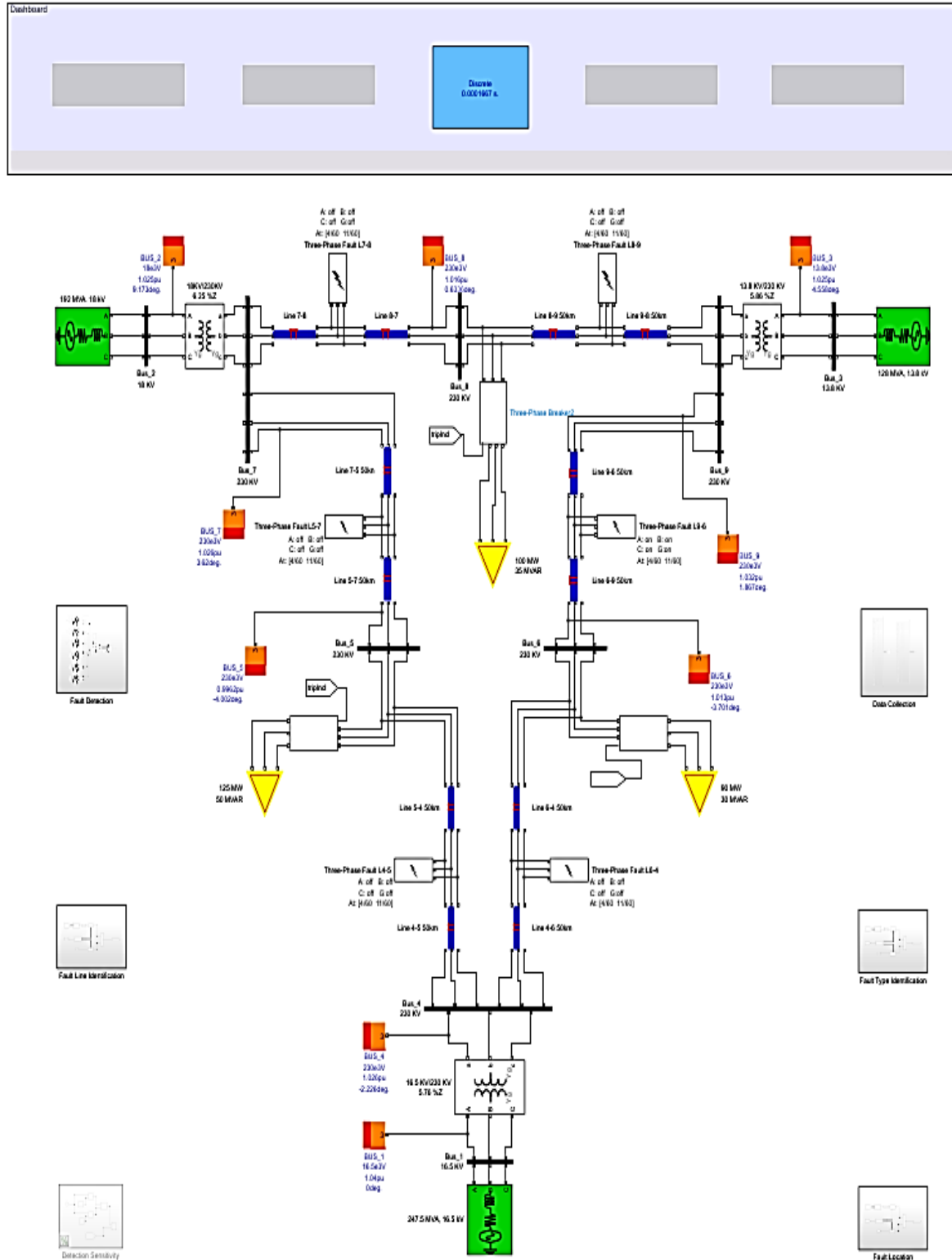


Figure 3.8: Modified IEEE 9 Bus Transmission System Used for Research

3.4.1.5 Fault Information Display

Upon the occurrence of a fault, information about the fault which has occurred are shown in a dashboard at the top of the Simulink model.

The dashboard contains 5 cards for displaying values. The first card displays “Faulted” when there is a fault and goes blank when there is no fault. The second card displays the transmission line along which the fault has occurred when there is a fault. The third card shows the powergui which is the MATLAB SIMULINK element to compute the simulations numerically. The fourth card shows the type of fault if there is a fault. Lastly, upon the occurrence of a fault, the fifth card shows the location of the fault along the transmission line where it occurred.

A typical scenario of when a BCG fault occurs along transmission line 7-8 at 64.2593 km from bus 7 is shown in figure 3.9 below.

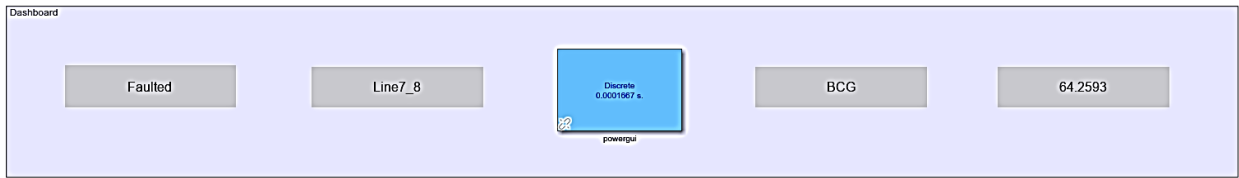


Figure 3.9: Dashboard Showing Information about Fault

3.4.2 Data Collection

The extracted features used for the processes of fault detection, fault line identification, fault type identification and fault locations are the moving averages and moving variances of the voltages at the six buses in the transmission system i.e. bus 4 to bus 9. Each bus has three lines, the total number of features is

$$\begin{aligned} \text{Number of features} &= 2 \text{ indicators} \times 3 \text{ Lines} \times 6 \text{ buses} \\ &= 36 \text{ Features} \end{aligned}$$

These features are listed in table 3.1 below:

S/N	Tag	Feature	S/N	Tag	Feature
1	MvgAvg_B4LA	Moving average of bus 4, line A Voltage	2	MvgAvg_B4LB	Moving average of bus 4, line B Voltage
3	MvgAvg_B4LC	Moving average of bus 4, line C Voltage	4	MvgAvg_B5LA	Moving average of bus 5, line A Voltage
5	MvgAvg_B5LB	Moving average of bus 5, line B Voltage	6	MvgAvg_B5LC	Moving average of bus 5, line C Voltage
7	MvgAvg_B6LA	Moving average of bus 6, line A Voltage	8	MvgAvg_B6LB	Moving average of bus 6, line B Voltage
9	MvgAvg_B6LC	Moving average of bus 6, line C Voltage	10	MvgAvg_B7LA	Moving average of bus 7, line A Voltage
11	MvgAvg_B7LB	Moving average of bus 7, line B Voltage	12	MvgAvg_B7LC	Moving average of bus 7, line C Voltage
13	MvgAvg_B8LA	Moving average of bus 8, line A Voltage	14	MvgAvg_B8LB	Moving average of bus 8, line B Voltage
15	MvgAvg_B8LC	Moving average of bus 8, line C Voltage	16	MvgAvg_B9LA	Moving average of bus 9, line A Voltage
17	MvgAvg_B9LB	Moving average of bus 9, line B Voltage	18	MvgAvg_B9LC	Moving average of bus 9, line C Voltage
19	MvgVar_B4LA	Moving variance of bus 4, line A Voltage	20	MvgVar_B4LB	Moving variance of bus 4, line B Voltage

21	MvgVar_B4LC	Moving variance of bus 4, line C Voltage	22	MvgVar_B5LA	Moving variance of bus 5, line A Voltage
23	MvgVar_B5LB	Moving variance of bus 5, line B Voltage	24	MvgVar_B5LC	Moving variance of bus 5, line C Voltage
25	MvgVar_B6LA	Moving variance of bus 6, line A Voltage	26	MvgVar_B6LB	Moving variance of bus 6, line B Voltage
27	MvgVar_B6LC	Moving variance of bus 6, line C Voltage	28	MvgVar_B7LA	Moving variance of bus 7, line A Voltage
29	MvgVar_B7LB	Moving variance of bus 7, line B Voltage	30	MvgAvg_B7LC	Moving variance of bus 7, line C Voltage
31	MvgVar_B8LA	Moving variance of bus 8, line A Voltage	32	MvgVar_B8LB	Moving variance of bus 8, line B Voltage
33	MvgVar_B8LC	Moving variance of bus 8, line C Voltage	34	MvgVar_B9LA	Moving variance of bus 9, line A Voltage
35	MvgVar_B9LB	Moving variance of bus 9, line B Voltage	36	MvgVar_B9LC	Moving variance of bus 9, line C Voltage

Table 3.1: Features used in the Approach Presented in this Project

Each of the 11 types of short-circuit faults is simulated at 21 different values of distances for each of the 6 transmission lines. The data generated from each combination of transmission line, fault type and distance were saved as a CSV file. A total of 1386 CSV files were generated. In addition to the 36 features above, each CSV file contains the fault indication i.e. whether a fault has been detected or not, the transmission line along which the fault has occurred, the type of fault and the distance along the transmission line where the fault has occurred.

3.5 Fault Detection

The most important aspect out of the 4 areas of research in this project is the fault detection aspect. Before line of fault identification, type of fault identification and fault location can be done, a fault must first be detected. A relay is attached to the fault detection system which switches of power in all the 3 load locations once a fault is detected.

A subsystem was created in the simulation model for the process of fault detection as shown in the figure 3.10 below.

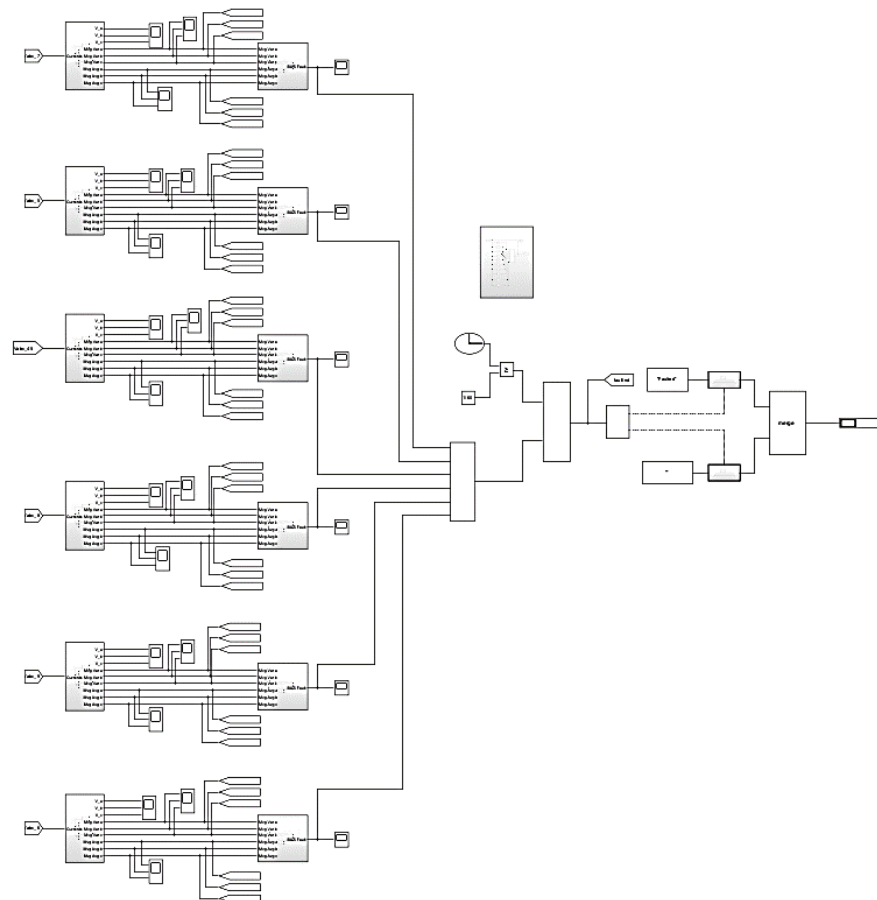


Figure 3.10: Fault Detection Subsystem in Simulink

In the subsystem, each fault voltage is checked for fault. If a fault is detected in any of the buses, the fault indication will be activated. The detection of fault for each bus is shown in figure 3.11 below.

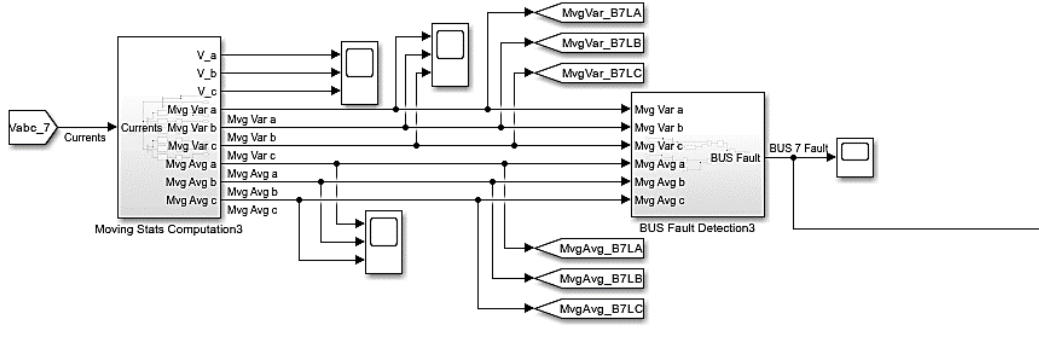


Figure 3.11: Fault Detection Process for each Bus

The process consists of two other subprocesses. The first one being the computation of moving statistics i.e. moving variance and moving average. The second step is using the moving statistics to detect the occurrence of fault. The process of calculating the moving statistics is shown in the figure 3.12 below.

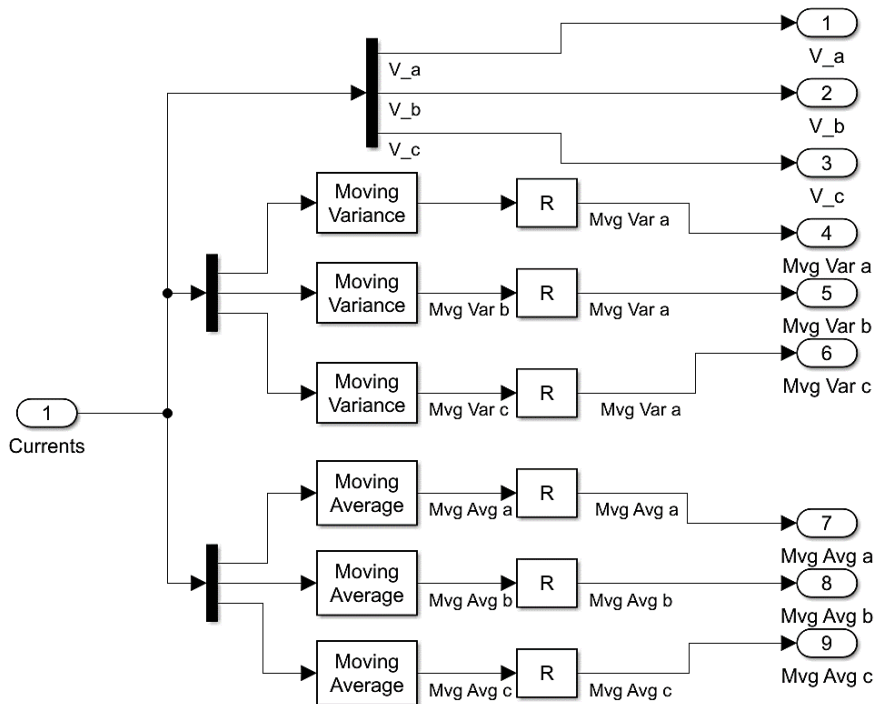


Figure 3.12: Process of Calculating Moving Statistics

The process of detecting fault using the calculated moving statistics from above is shown in figure 3.13 below.

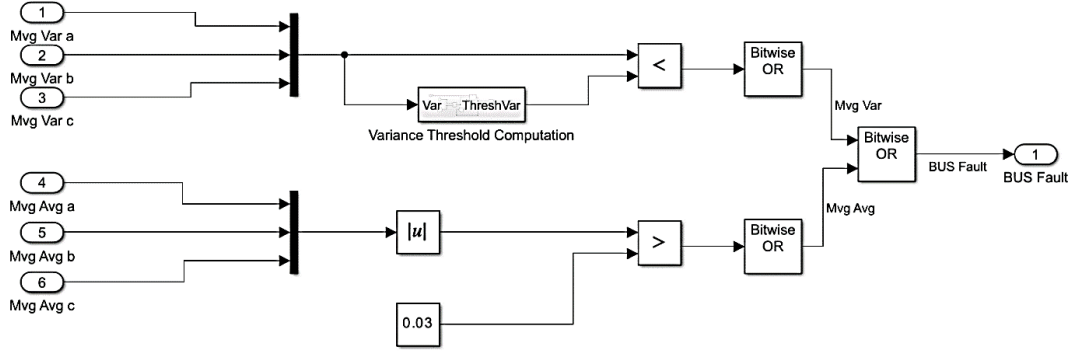


Figure 3.13: Fault Detection Using the Moving Statistics

3.6 Fault Line Identification

After the fault identification process has been completed. A decision tree classification model was created making use of the 36 line features in section 3.4.2 as the predictor variables and the transmission line along which fault occurs as the response variable. The Fault Line Identification subsystem is shown in figure 3.14 below.

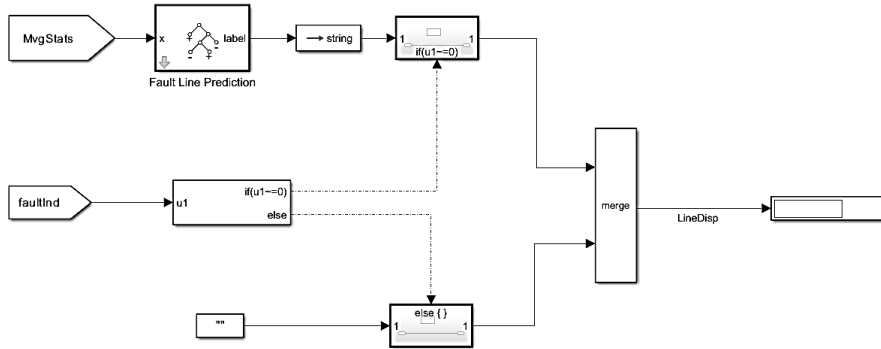


Figure 3.14: Fault Line Identification Subsystem

3.7 Fault Type Identification

A decision tree classification model was also built for the identification of the type of fault that has occurred. The predictor variables are the 36 line features as done for fault line identification. However, in this case, the response variable used to build the model is the fault type. The Fault Type Identification subsystem is shown in figure 3.15 below.

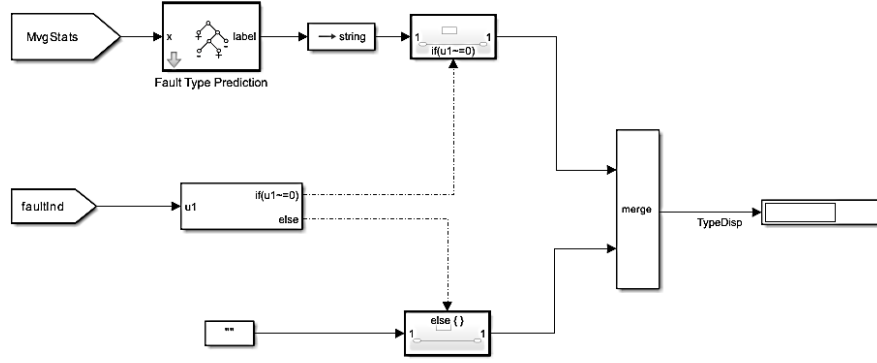


Figure 3.15: Fault Type Identification

3.8 Fault Location

Finally, a decision tree regression model was built for the prediction of where a fault has occurred along the transmission line. The predictor variables of the fault location model are the same as the two models above. The predictor variables are the 36 features in section 3.4.2. However, in this case, the response variable used to build the model is the fault location which is a continuous value unlike the response variable of the two models above which are categorical. The Fault Location subsystem is shown in figure 3.16 below.

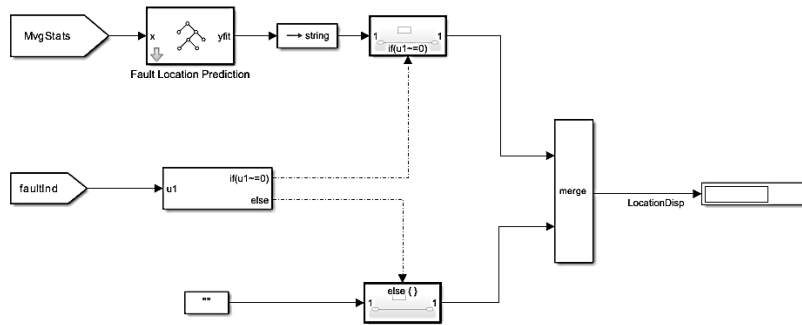


Figure 3.16: Fault Location Subsystem

3.9 Relay System

A relay system is implemented to protect the consumers from the faults. The three-phase circuit breakers added are used to disconnect the loads (distribution system) from the transmission system when a fault is detected.

In the relay system, ten consecutive values of the fault indicator are tracked. If all ten consecutive indicators are true, then the relays would be activated. The reason for counting ten consecutive values is to ensure that the fault line identification model, fault type identification model and the fault location models have enough inputs to make accurate predictions. The interval between each value calculation is $(1/6) \times 10^{-3}$ which is approximately 0.0001667 seconds or 0.1667 milliseconds as shown in the powergui solver for the simulation.

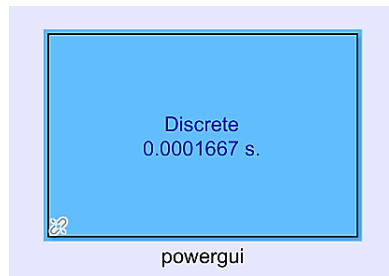


Figure 3.17: Simulation Powergui

The time taken to activate relay after fault detection is therefore the interval between ten consecutive values as calculated below.

$$\text{Relay Activation Time} = (10 - 1) \times \text{values interval}$$

$$\text{Relay Activation Time} = (10 - 1) \times \frac{1}{6} \times 10^{-3}$$

$$\text{Relay Activation Time} = 0.0015 \text{ seconds}$$

$$\text{Relay Activation Time} = 1.5 \text{ milliseconds}$$

3.10 Block Diagram

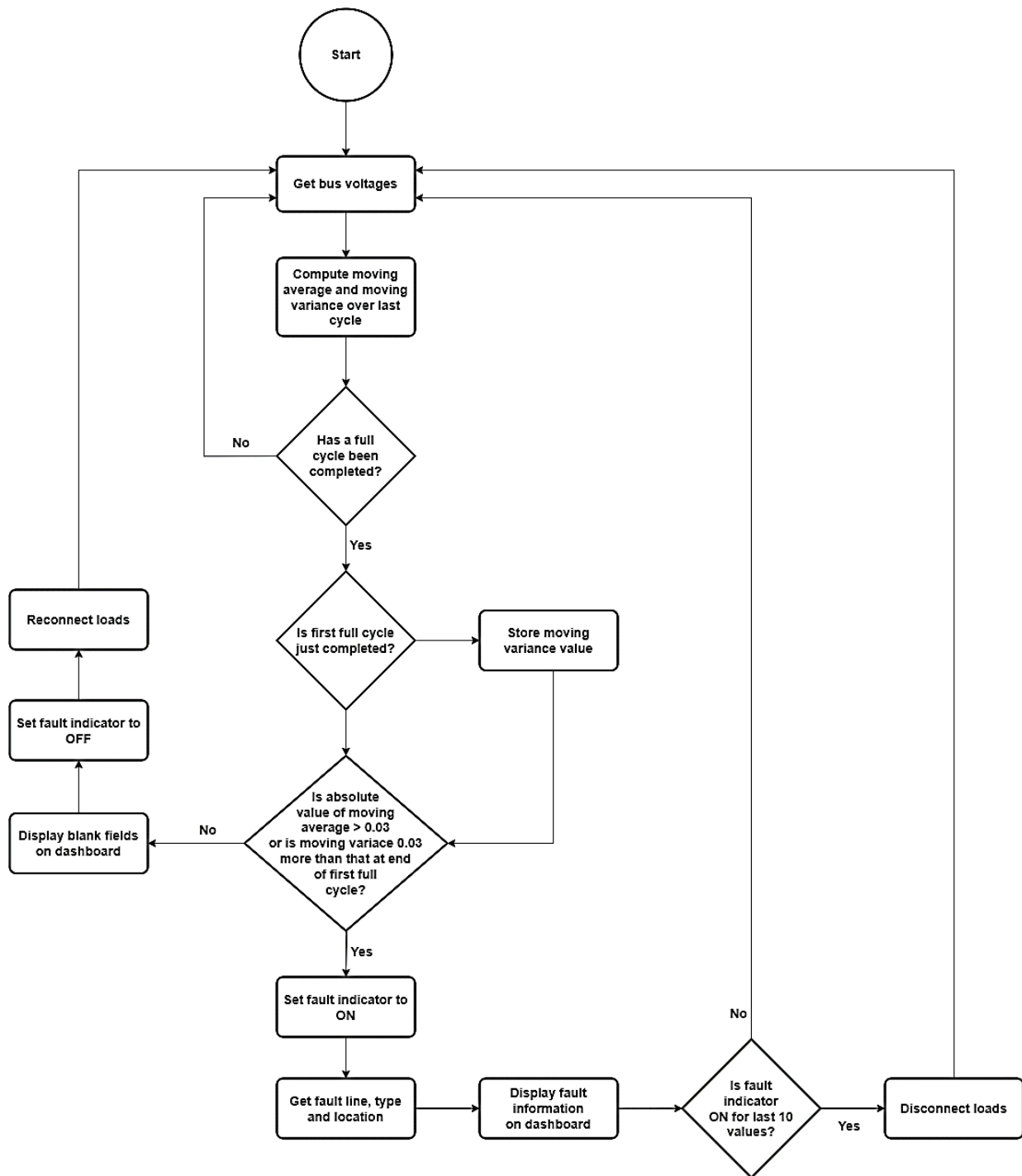


Figure 3.18: Block Diagram showing the processes in the proposed approach

CHAPTER FOUR: RESULTS AND ANALYSIS

4.1 Introduction

The results of this project are provided in this chapter based on the five objectives set for this project. Simulations were done in MATLAB Simulink making use of the modified IEEE 9 bus system to assess the effectiveness of the proposed methodologies.

4.2 Fault Detection

The effectiveness of the fault detection phase is assessed based on the detection time i.e. how fast can faults be detected upon occurrence.

The fault detection behaviour of this approach when an AB fault occurs along transmission line 7-8 at 50km from bus 7 is shown in figure 4.1 below.

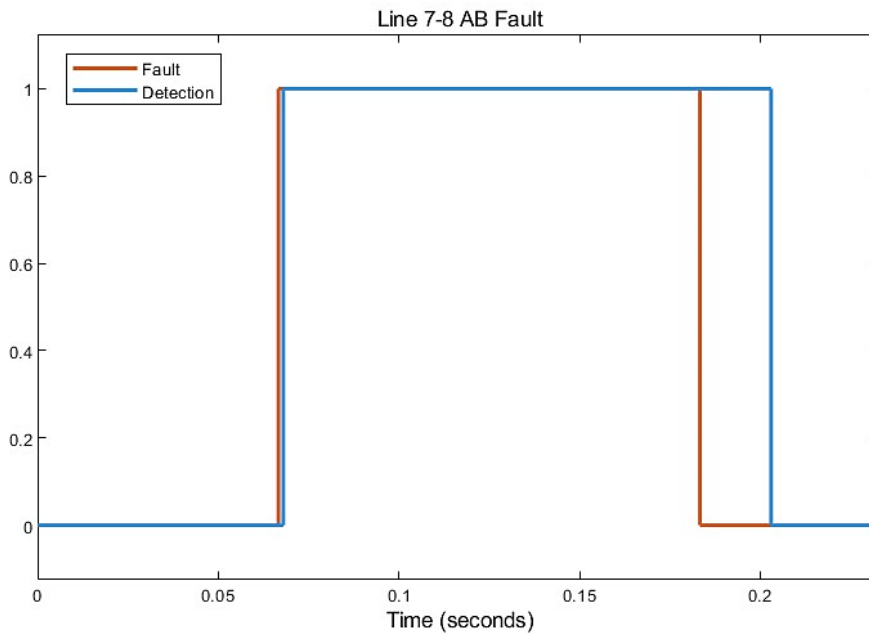


Figure 4.1: Fault Occurrence and Detection of AB Fault on Line 7-8 at 50km from Bus 7

From the figure above, it could be seen that the faults were detected almost instantaneously. However, when the fault stops the detection still continue for a small period of time. This is as a result of the transient effect of the fault which does not die away immediately the fault stops.

For simplicity of the results, the results are shown for faults occurring at the middle of the transmission line i.e. at 50km from both ends of the transmission lines. The fault detection behaviours of each fault along each transmission line are now summarized in the following sections.

4.2.1 Fault Detection on Line 7-8

The time between fault occurrences and fault detections (detection time) in milliseconds for faults on transmission line 7-8 are summarized in table 4.1 below.

Fault	Fault Occurrence Time (milliseconds)	Fault Detection Time (milliseconds)	Detection Time (milliseconds)
AG	66.6667	69.8333	3.17
BG	66.6667	67.5000	0.83
CG	66.6667	67.3333	0.67
AB	66.6667	68.0000	1.33
AC	66.6667	67.5000	0.83
BC	66.6667	67.3333	0.67
ABG	66.6667	67.5000	0.83
ACG	66.6667	67.3333	0.67
BCG	66.6667	67.3333	0.67
ABC	66.6667	67.3333	0.67
ABCG	66.6667	67.3333	0.67

Table 4.1: Fault Detection Times Per Type of Fault on Transmission Line 7-8

From the results above, the slowest detection on line 7-8 took 3.17 milliseconds after fault occurrence. While the fastest detection took 0.67 milliseconds.

4.2.2 Fault Detection on Line 8-9

The time between fault occurrences and fault detections (detection time) in milliseconds for faults on transmission line 8-9 are summarized in table 4.2 below.

Fault	Fault Occurrence Time (milliseconds)	Fault Detection Time (milliseconds)	Detection Time (milliseconds)
AG	66.6667	70.0000	3.33
BG	66.6667	67.6667	1.00
CG	66.6667	67.5000	0.83
AB	66.6667	68.1667	1.50

AC	66.6667	67.5000	0.83
BC	66.6667	67.5000	0.83
ABG	66.6667	67.6667	1.00
ACG	66.6667	67.5000	0.83
BCG	66.6667	67.5000	0.83
ABC	66.6667	67.3333	0.67
ABCG	66.6667	67.3333	0.67

Table 4.2: Fault Detection Times Per Type of Fault on Transmission Line 8-9

From the results above, the slowest detection on line 8-9 took 3.33 milliseconds after fault occurrence. While the fastest detection took 0.67 milliseconds.

4.2.3 Fault Detection on Line 9-6

The time between fault occurrences and fault detections (detection time) in milliseconds for faults on transmission line 9-6 are summarized in table 4.3 below.

Fault	Fault Occurrence Time (milliseconds)	Fault Detection Time (milliseconds)	Detection Time (milliseconds)
AG	66.6667	70.1667	3.50
BG	66.6667	67.8333	1.17
CG	66.6667	67.5000	0.83
AB	66.6667	68.3333	1.67
AC	66.6667	67.6667	1.00
BC	66.6667	67.5000	0.83
ABG	66.6667	67.8333	1.17
ACG	66.6667	67.5000	0.83
BCG	66.6667	67.5000	0.83
ABC	66.6667	67.5000	0.83
ABCG	66.6667	67.5000	0.83

Table 4.3: Fault Detection Times Per Type of Fault on Transmission Line 9-6

From the results above, the slowest detection on line 9-6 took 3.50 milliseconds after fault occurrence. While the fastest detection took 0.83 milliseconds.

4.2.4 Fault Detection on Line 6-4

The time between fault occurrences and fault detections (detection time) in milliseconds for faults on transmission line 6-4 are summarized in table 4.4 below.

Fault	Fault Occurrence Time (milliseconds)	Fault Detection Time (milliseconds)	Detection Time (milliseconds)
AG	66.6667	69.8333	3.17
BG	66.6667	67.6667	1.00
CG	66.6667	67.3333	0.67
AB	66.6667	68.0000	1.33
AC	66.6667	67.5000	0.83
BC	66.6667	67.3333	0.67
ABG	66.6667	67.6667	1.00
ACG	66.6667	67.3333	0.67
BCG	66.6667	67.3333	0.67
ABC	66.6667	67.3333	0.67
ABCG	66.6667	67.3333	0.67

Table 4.4: Fault Detection Times Per Type of Fault on Transmission Line 6-4

From the results above, the slowest detection on line 6-4 took 3.17 milliseconds after fault occurrence. While the fastest detection took 0.67 milliseconds.

4.2.5 Fault Detection on Line 4-5

The time between fault occurrences and fault detections (detection time) in milliseconds for faults on transmission line 4-5 are summarized in table 4.5 below.

Fault	Fault Occurrence Time (milliseconds)	Fault Detection Time (milliseconds)	Detection Time (milliseconds)
AG	66.6667	69.6667	3.00
BG	66.6667	67.5000	0.83
CG	66.6667	67.3333	0.67
AB	66.6667	68.0000	1.33

AC	66.6667	67.5000	0.83
BC	66.6667	67.3333	0.67
ABG	66.6667	67.5000	0.83
ACG	66.6667	67.3333	0.67
BCG	66.6667	67.3333	0.67
ABC	66.6667	67.3333	0.67
ABCG	66.6667	67.3333	0.67

Table 4.5: Fault Detection Times Per Type of Fault on Transmission Line 4-5

From the results above, the slowest detection on line 4-5 took 3.00 milliseconds after fault occurrence. While the fastest detection took 0.67 milliseconds.

4.2.6 Fault Detection on Line 5-7

The time between fault occurrences and fault detections (detection time) in milliseconds for faults on transmission line 5-7 are summarized in table 4.6 below.

Fault	Fault Occurrence Time (milliseconds)	Fault Detection Time (milliseconds)	Detection Time (milliseconds)
AG	66.6667	70.0000	3.33
BG	66.6667	67.6667	1.00
CG	66.6667	67.5000	0.83
AB	66.6667	68.1667	1.50
AC	66.6667	67.6667	1.00
BC	66.6667	67.5000	0.83
ABG	66.6667	67.6667	1.00
ACG	66.6667	67.5000	0.83
BCG	66.6667	67.5000	0.83
ABC	66.6667	67.5000	0.83
ABCG	66.6667	67.5000	0.83

Table 4.6: Fault Detection Times Per Type of Fault on Transmission Line 5-7

From the results above, the slowest detection on line 5-7 took 3.33 milliseconds after fault occurrence. While the fastest detection took 0.83 milliseconds.

4.2.7 Overall Slowest and Fastest Fault Detection Time

The overall slowest fault detection time observed using the approach presented in this project is 3.50 milliseconds while the fastest detection time took 0.67 milliseconds.

4.3 Transmission Line Identification

As explained in section 3.6, a decision tree model was created to predict the transmission line where a fault has occurred based on the 36 features explained in section 3.4.2. The transmission line prediction would only be shown in the dashboard explained in section 3.4.1.5 after a fault has been detected. When a fault has not been detected, the output will be blank in the dashboard.

Figure 4.2 is a confusion matrix showing the performance of the transmission line identification model.

Confusion Matrix for Line Identification
Overall Accuracy = 99.2075%

True Class	Line 4-5	98.9%	0.5%	0.5%		0.0%	0.0%
	Line 5-7	0.5%	99.4%	0.0%	0.0%	0.0%	0.0%
	Line 6-4	0.7%	0.0%	98.8%	0.0%	0.0%	0.5%
	Line 7-8	0.0%	0.0%	0.0%	99.9%	0.0%	
	Line 8-9	0.0%	0.0%	0.0%	0.0%	99.3%	0.6%
	Line 9-6	0.0%	0.0%	0.3%	0.0%	0.6%	99.0%
		Line 4-5	Line 5-7	Line 6-4	Line 7-8	Line 8-9	Line 9-6
		Predicted Class					

Figure 4.2: Performance of Fault Transmission Line Identification Model

4.4 Fault Type Identification

A decision tree model was built as explained in section 3.7 to predict the type of fault that has occurred making use of the 36 extracted features of the transmission system explained in section 3.4.2. Just like the fault line identification model, the model prediction would only be displayed in the dashboard explained in section 3.4.1.5 when a fault has been detected. When a fault has not been detected, the output will be blank in the dashboard.

The performance of the fault type identification model is shown in the confusion matrix in figure 4.3 below. Both ABC and ABCG faults are categorized as a symmetrical fault, “symFault” because the effects of these two faults are identical.

Confusion Matrix for Line Identification
Overall Accuracy = 99.9764%

AB	100.0%	0.0%								0.0%
ABG		100.0%						0.0%		0.0%
AC			100.0%							
ACG		0.0%		100.0%	0.0%				0.0%	0.0%
AG		0.0%		0.0%	100.0%					0.0%
BC						100.0%	0.0%		0.0%	
BCG		0.0%					100.0%		0.0%	0.0%
BG		0.0%						100.0%		0.0%
CG							0.0%		100.0%	
symFault		0.0%		0.0%	0.0%		0.0%			100.0%
	AB	ABG	AC	ACG	AG	BC	BCG	BG	CG	symFault

Predicted Class

Figure 4.3: Performance of Fault Type Identification Model

4.5 Fault Location

A decision tree regression model was to predict the location (in kilometers) along the transmission line where a fault has occurred as explained in section 3.8. The features used for this model are the 36 features extracted from the transmission system as explained in section 3.4.2. The output of this model would only be displayed in the dashboard explained in section 3.4.1.5 when a fault has been detected. When a fault has not been detected the output would be blank.

The performance of the fault location model was evaluated using the mean squared error of the prediction on the test dataset which is a hold-out set. The test dataset is 20% of the full dataset.

The mean squared error of this model on the test dataset is 24.4096.

4.6 Relay System

As explained in section 3.9, a relay system is triggered 1.5 milliseconds after a fault has been detected to safeguard the loads (distribution systems) attached to the transmission system.

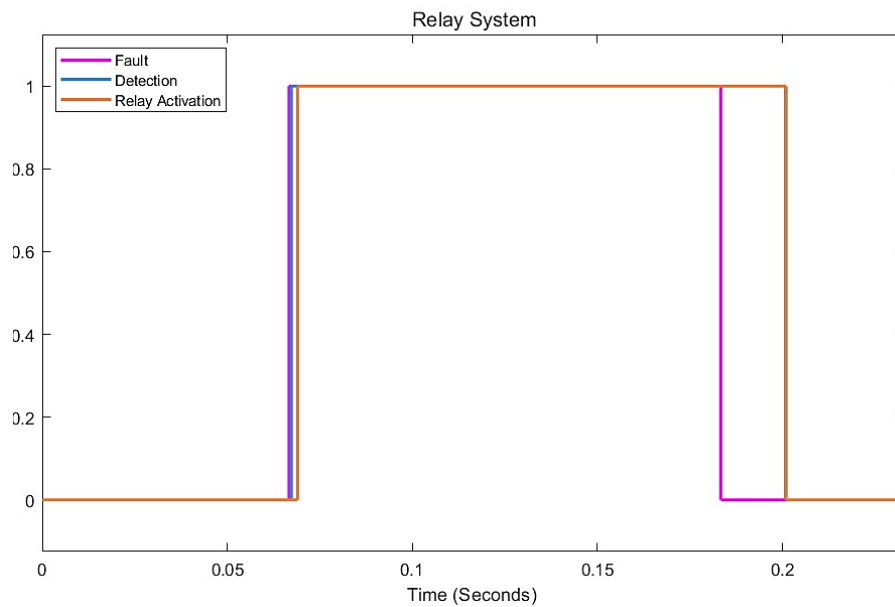


Figure 4.4: Relay System Activation

However, immediately the fault detection indicator goes back to zero showing that there is no more fault, the load would be immediately reconnected. The fault detection indicator goes back to zero few milliseconds after the fault has disappeared from the transmission system.

CHAPTER FIVE: SUMMARY, CONCLUSION AND RECOMMENDATION

5.1 Summary

From the results in the previous chapters, the following discussions arise to better summarize and interpret the results.

5.1.1 Overall Distribution of Detection Times

The count of detection times is shown in figure 5.1 below.

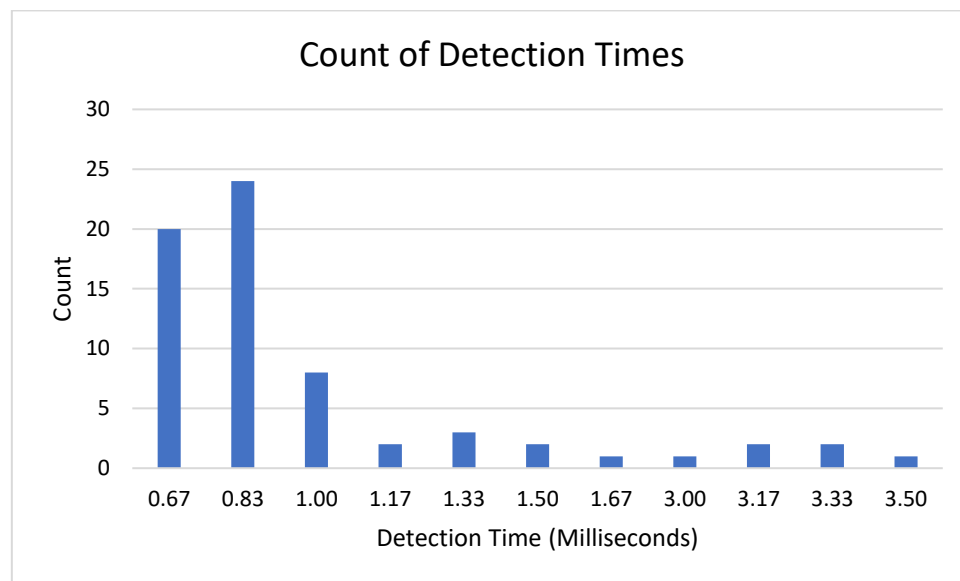


Figure 5.1: Count of Fault Detection Times

From the figure above, the most probable time to detect a fault is 0.83 milliseconds followed by 0.67 milliseconds then followed by 1.00 milliseconds and so on.

The distribution of the detection time is shown in figure 5.2 below.

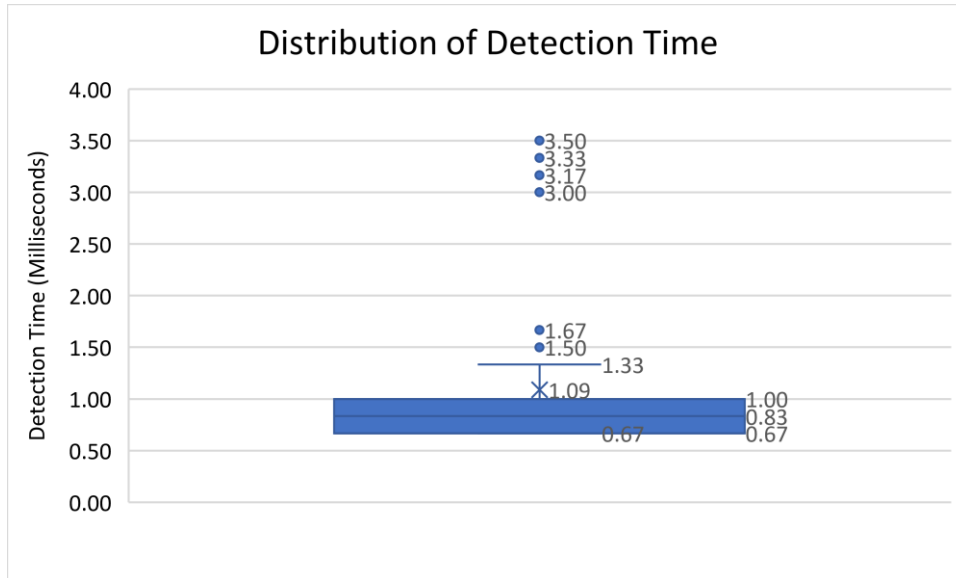


Figure 5.2: Distribution of Fault Detection Time

From the analysis of fault detection time shown in figure 5.2 above, the following statistics are gotten.

Statistic	Value (Milliseconds)
Minimum	0.67
25th Percentile	0.67
Median	0.83
75th Percentile	1.00
Mean	1.09
Upper Outlier Cut-off	1.33
Maximum	3.50

Table 5.1: Statistics from the Distribution of Fault Detection Time

These results can be interpreted as:

1. The lowest value (fastest fault detection time) possible is 0.67 milliseconds.
2. A fault should be expected to take 0.83 milliseconds (based on the median value) or 1.09 milliseconds (based on the mean value) to be detected.
3. Majority of the faults should be expected to be detected within 0.67 milliseconds (25th percentile) and 1.00 milliseconds (75th percentile).
4. Fault detection time of more than 1.33 milliseconds are very rare.

5. Fault detection time should be expected to not exceed 3.50 milliseconds in a normal circumstance.

5.1.2 Distribution of Detection Time Based on Line of Fault

It was discovered that fault detection times along the transmission lines are relatively faster than one another. The distribution of detection time based on the line of transmission where the fault has occurred is shown in figure 5.3 below.

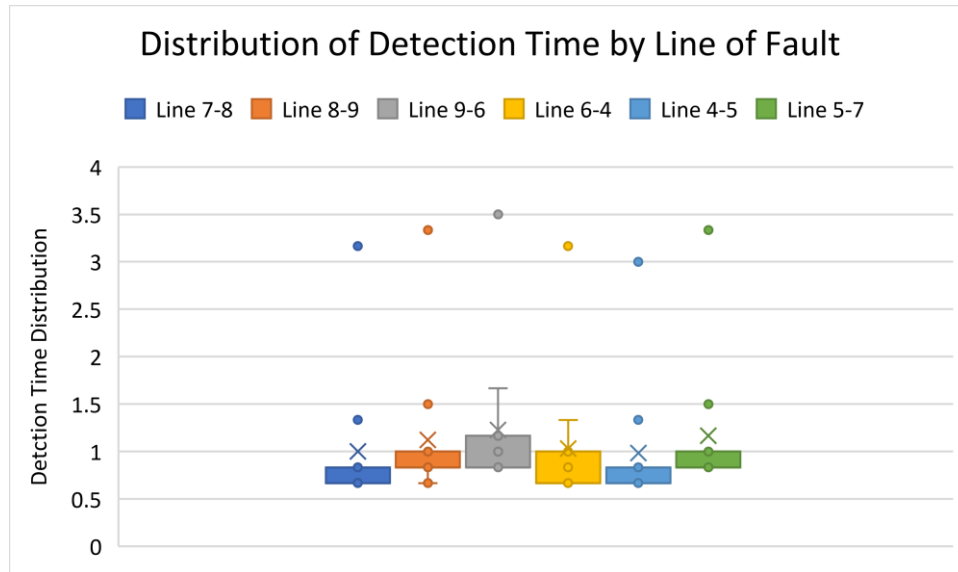


Figure 5.3: Distribution of Fault Detection Times Based on Transmission Line of Occurrence

From the distribution above, the following interpretation can be gotten:

1. Faults along Transmission line 7-8 typically take between 0.67 milliseconds (25th percentile) to 0.83 milliseconds (75th percentile) but can be up to 3.17 milliseconds (maximum).
2. Faults along Transmission line 8-9 typically take between 0.83 milliseconds (25th percentile) to 1.00 milliseconds (75th percentile) but can be as fast as 0.67 milliseconds (minimum) and as slow as 3.33 milliseconds (maximum).
3. Faults along Transmission line 9-6 typically take between 0.83 milliseconds (25th percentile) to 1.23 milliseconds (75th percentile) but can be up to 3.50 milliseconds (maximum).
4. Faults along Transmission line 6-4 typically take between 0.67 milliseconds (25th percentile) to 1.00 milliseconds (75th percentile) but can be up to 3.17 milliseconds (maximum).

5. Faults along Transmission line 4-5 typically take between 0.67 milliseconds (25th percentile) to 0.83 milliseconds (75th percentile) but can be up to 3.00 milliseconds (maximum).
6. Faults along Transmission line 5-7 typically take between 0.83 milliseconds (25th percentile) to 1.00 milliseconds (75th percentile) but can be up to 3.33 milliseconds (maximum).

5.1.3 Distribution of Detection Time Based on Type of Fault

It was also discovered that some types of fault take relative longer time than others to be detected. The distribution of detection time based on the type of fault is shown in figure 5.4 below.

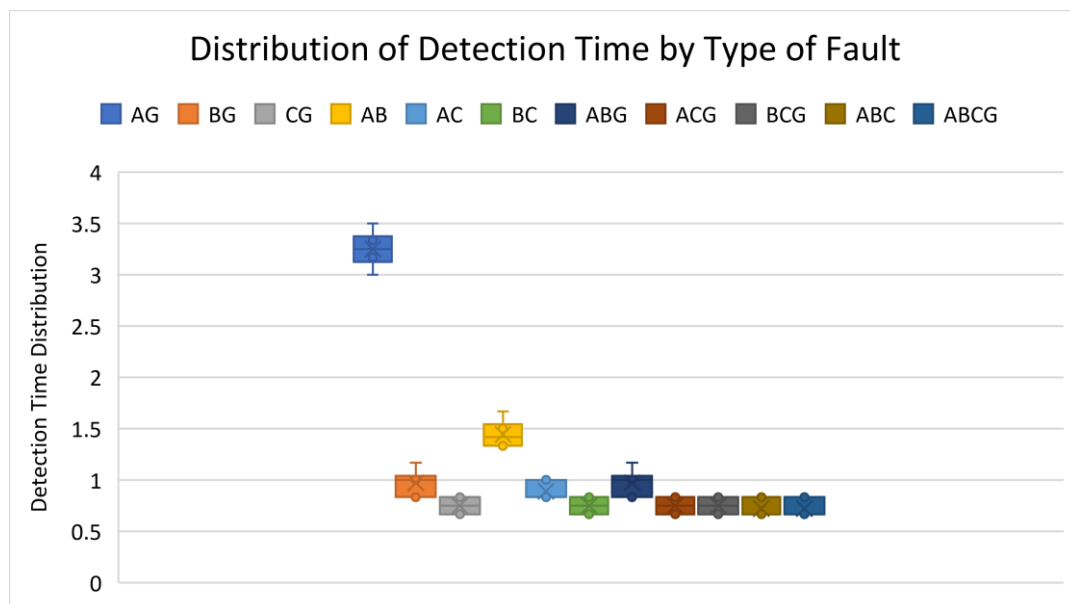


Figure 5.4: Distribution of Fault Detection Times Based on Type of Fault

From the distribution shown above, it could be seen that AG faults takes much more time to be detected relative to other types of fault. The following interpretation can be given:

1. AG faults typically take between 3.17 milliseconds (25th percentile) to 3.33 milliseconds to be detected but can be as fast as 3.00 milliseconds and as slow as 3.50 milliseconds.
2. BG faults typically take between 0.83 milliseconds (25th percentile) to 1.04 milliseconds to be detected but can be as slow as 1.17 milliseconds.
3. CG faults typically take between 0.67 milliseconds (25th percentile) to 0.83 milliseconds to be detected.

4. AB faults typically take between 1.33 milliseconds (25th percentile) to 1.54 milliseconds to be detected but can be as slow as 1.67 milliseconds.
5. AC faults typically take between 0.83 milliseconds (25th percentile) to 1.00 milliseconds to be detected.
6. BC faults typically take between 0.67 milliseconds (25th percentile) to 0.83 milliseconds to be detected.
7. ABG faults typically take between 0.83 milliseconds (25th percentile) to 1.04 milliseconds to be detected but can be as slow as 1.17 milliseconds.
8. ACG faults typically take between 0.67 milliseconds (25th percentile) to 0.83 milliseconds to be detected.
9. BCG faults typically take between 0.67 milliseconds (25th percentile) to 0.83 milliseconds to be detected.
10. ABC faults typically take between 0.67 milliseconds (25th percentile) to 0.83 milliseconds to be detected.
11. BG faults typically take between 0.67 milliseconds (25th percentile) to 0.83 milliseconds to be detected.

5.2 Conclusion

In conclusion, a new approach for detecting faults along transmission lines has been presented and validated through the simulation of a model of the IEEE 9 bus transmission system in MATLAB SIMULINK.

The detection time based on this approach is typically between 0.67 milliseconds and 1.00 milliseconds but can be up to 3.50 milliseconds.

Furthermore, three machine learning models were implemented to identify the transmission line along which a fault has occurred, the type of fault that has occurred and the location along the transmission line where the fault has occurred upon the occurrence of faults.

With the adoption of these approach for fault detection, classification and location along transmission lines, the time and effort taken to restore power after a fault has occurred can drastically be reduce which would result in increased sustainability and reliability of the electrical grid and also more income for power transmission companies.

5.3 Recommendation

Further research can be done in continuation of this project in the following areas:

1. Assessment of the effectiveness of various machine learning models in predicting the transmission line along which faults have occurred based on the bus voltages.
2. Assessment of the effectiveness of various machine learning models in predicting the type of fault which has occurred based on the bus voltages.
3. Assessment of the effectiveness of various machine learning models in predicting the location along the transmission line where a fault has occurred.
4. Implementation of the proposed approach in power transmission monitoring systems and relay systems.

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APPENDIX A: DATASET GENERATION CODE

Code for generating dataset from the MATLAB SIMULINK model. This code generates 1386 CSV files. This code typically takes more than 12 hours to generate the CSV files.

```
% specify the transmission lines in the transmission network
faultLineArray = ["7-8" "8-9" "9-6" "6-4" "4-5" "5-7"];

% specify types of faults
faultTypeArray = ["ABCG" "ABC" "ABG" "ACG" "BCG" "AG" "BG" "CG" "AB" "AC" "BC"];

% set fault duration
faultDuration = '[4/60 11/60]';

% open the Simulink model
open_system('IEEE_9bus_new_o');

% switch off all faults for all transmission lines
for faultLine = faultLineArray
    faultBlock = strjoin(["IEEE_9bus_new_o/Three-Phase Fault L" faultLine], "");
    set_param(faultBlock, 'FaultA', 'off');
    set_param(faultBlock, 'FaultB', 'off');
    set_param(faultBlock, 'FaultC', 'off');
    set_param(faultBlock, 'GroundFault', 'off');
end

% for each transmission line
for faultLine = faultLineArray
    % select fault block for current transmission line
    faultBlock = strjoin(["IEEE_9bus_new_o/Three-Phase Fault L" faultLine], "");
    % set duration of fault for this block
    set_param(faultBlock, 'SwitchTimes', faultDuration)

    % for each type of fault
    for faultType = faultTypeArray
        % set the type of fault to the current one
        switch faultType
            case "ABCG"
                set_param(faultBlock, 'FaultA', 'on');
                set_param(faultBlock, 'FaultB', 'on');
```

```

    set_param(faultBlock, 'FaultC', 'on');
    set_param(faultBlock, 'GroundFault', 'on');
case "ABC"
    set_param(faultBlock, 'FaultA', 'on');
    set_param(faultBlock, 'FaultB', 'on');
    set_param(faultBlock, 'FaultC', 'on');
    set_param(faultBlock, 'GroundFault', 'off');
case "ABG"
    set_param(faultBlock, 'FaultA', 'on');
    set_param(faultBlock, 'FaultB', 'on');
    set_param(faultBlock, 'FaultC', 'off');
    set_param(faultBlock, 'GroundFault', 'on');
case "ACG"
    set_param(faultBlock, 'FaultA', 'on');
    set_param(faultBlock, 'FaultB', 'off');
    set_param(faultBlock, 'FaultC', 'on');
    set_param(faultBlock, 'GroundFault', 'on');
case "BCG"
    set_param(faultBlock, 'FaultA', 'off');
    set_param(faultBlock, 'FaultB', 'on');
    set_param(faultBlock, 'FaultC', 'on');
    set_param(faultBlock, 'GroundFault', 'on');
case "AG"
    set_param(faultBlock, 'FaultA', 'on');
    set_param(faultBlock, 'FaultB', 'off');
    set_param(faultBlock, 'FaultC', 'off');
    set_param(faultBlock, 'GroundFault', 'on');
case "BG"
    set_param(faultBlock, 'FaultA', 'off');
    set_param(faultBlock, 'FaultB', 'on');
    set_param(faultBlock, 'FaultC', 'off');
    set_param(faultBlock, 'GroundFault', 'on');
case "CG"
    set_param(faultBlock, 'FaultA', 'off');
    set_param(faultBlock, 'FaultB', 'off');
    set_param(faultBlock, 'FaultC', 'on');
    set_param(faultBlock, 'GroundFault', 'on');
case "AB"
    set_param(faultBlock, 'FaultA', 'on');
    set_param(faultBlock, 'FaultB', 'on');

```

```

        set_param(faultBlock, 'FaultC', 'off');
        set_param(faultBlock, 'GroundFault', 'off');
    case "AC"
        set_param(faultBlock, 'FaultA', 'on');
        set_param(faultBlock, 'FaultB', 'off');
        set_param(faultBlock, 'FaultC', 'on');
        set_param(faultBlock, 'GroundFault', 'off');
    case "BC"
        set_param(faultBlock, 'FaultA', 'off');
        set_param(faultBlock, 'FaultB', 'on');
        set_param(faultBlock, 'FaultC', 'on');
        set_param(faultBlock, 'GroundFault', 'off');
end

% set the line of fault string to add to the generated data
faultLineStr = strjoin(["Line", faultLine]);

% specify the fault location between 0.1km and 9.9km
for faultLength = [0.1 5:5:99 99.9]
    % set the current fault location
    eval(strjoin(["L" strep(faultLine, "-", "") "_Length = " ...
        string(faultLength) ";"], ""));

    % set the location of fault string to add to the generated data
    faultDistance = faultLength;

    % simulate the model with the modifications
    sim('IEEE_9bus_new_o.slx');

    % save the generated data for the current configurations to a
    % csv file
    saveToCSV(lineSignals, strjoin(["Datasets/L" strep(faultLine, "-",
"" ) ...
        "_ " string(faultType) "_ " strep(string(eval(strjoin(["L" ...
        strep(faultLine, "-", "") "_Length"], ""))), ".", "_")
".csv"], ""));
end
disp(strjoin(["Done generating data for Line" faultLine ...
    "and type" faultType]))
end

```

```

% switch off all faults for current transmission line
set_param(faultBlock, 'FaultA', 'off');
set_param(faultBlock, 'FaultB', 'off');
set_param(faultBlock, 'FaultC', 'off');
set_param(faultBlock, 'GroundFault', 'off');

% display message to show end of data generation for current line
disp(repmat('=',1,44))
disp(strjoin(["Done generating data for Line" faultLine]))
disp(repmat('=',1,44))
end

% make 2 beep sounds to signify end of operation
beep
pause(2)
beep

```

APPENDIX B: FULL DATASET GENERATION CODE

Code for combining the 1386 CSV files into 1 CSV file.

```

% datasets folder
datasetFolder = "Datasets";

% get a list of all csv files (datasets) in the folder
csvFiles = dir(fullfile(datasetFolder, 'L*.csv'));

% initialize an empty table for the combined data
fullDataset = [];

% loop through each csv file and append its data to fullDataset
for k = 1:length(csvFiles)
    %Get the full path of the current csv file
    filePath = fullfile(datasetFolder, csvFiles(k).name);

    % read the current csv file into a table
    data = readtable(filePath);

```

```

    % append the data to the combined table
    fullDataset = [fullDataset; data];
end

% specify the output file path for the combined csv file
combinedDataPath = fullfile(datasetFolder, 'full_dataset.csv');

% finally, write the combined data to the new csv file
writetable(fullDataset, combinedDataPath);

% write a message to notify end of operation
disp('All csv files have been merged into full_dataset.csv');

% make 2 beep sounds to signify end of operation
beep
pause(2)
beep

```

APPENDIX C: CODE FOR BUILDING LINE IDENTIFICATION MODEL

Code for building model to identify faulty transmission line.

```

% load the full dataset
fullDataset = readtable("Datasets\full_dataset.csv");

% define the predictors and the response data
predictorData = fullDataset(fullDataset.faultIndicator == 1,5:end);
responseData = fullDataset(fullDataset.faultIndicator == 1,{'faultLine'});
responseData = categorical(responseData{:,:});

% split data into training and test sets
cv = cvpartition(size(predictorData, 1), 'HoldOut', 0.2);
trainIndex = training(cv);
testIndex = test(cv);

trainPredictorData = predictorData(trainIndex, :);
trainResponseData = responseData(trainIndex, :);
testPredictorData = predictorData(testIndex, :);

```

```

testResponseData = responseData(testIndex, :);

% train a decision tree model on the dataset
treeModel = fitcecoc(trainPredictorData, trainResponseData);

% check the model performance
predictions = predict(treeModel, testPredictorData);
accuracy = sum(predictions == testResponseData)/(length(testResponseData));
disp(['Model Accuracy: ', num2str(accuracy * 100), '%']);

figure()
confusionchart(testResponseData, predictions, 'Normalization', ...
    'row-normalized');
titleString = ["Confusion Matrix for Line Identification";...
    strjoin(["Overall Accuracy = " num2str(accuracy*100) "%"], "")];
title(titleString);

% save the model
save('lineIdentificationModel_svm.mat', 'treeModel');

% make 2 beep sounds to signify end of operation
beep
pause(2)
beep

```

APPENDIX D: CODE FOR BUILDING FAULT TYPE IDENTIFICATION MODEL

Code for building model to identify type of fault.

```

% load the full dataset
fullDataset = readtable("Datasets\full_dataset.csv");

% combine label both ABC and ABCG faults as SymFault
symInd = (fullDataset.faultType == "ABCG" | fullDataset.faultType == "ABC");
fullDataset.faultType(symInd) = {'symFault'};

% define the predictors and the response data
predictorData = fullDataset(fullDataset.faultIndicator == 1,5:end);

```

```

responseData = fullDataset(fullDataset.faultIndicator == 1,{'faultType'});
responseData = categorical(responseData{:, :});

% split data into training and test sets
cv = cvpartition(size(predictorData, 1), 'HoldOut', 0.2);
trainIndex = training(cv);
testIndex = test(cv);

trainPredictorData = predictorData(trainIndex, :);
trainResponseData = responseData(trainIndex, :);
testPredictorData = predictorData(testIndex, :);
testResponseData = responseData(testIndex, :);

% train a decision tree model on the dataset
treeModel = fitctree(trainPredictorData, trainResponseData);

% check the model performance
predictions = predict(treeModel, testPredictorData);
accuracy = sum(predictions == testResponseData)/(length(testResponseData));
disp(['Model Accuracy: ', num2str(accuracy * 100), '%']);

confusionchart(testResponseData, predictions, 'Normalization', ...
    'row-normalized');
titleString = ["Confusion Matrix for Line Identification";...
    strjoin(["Overall Accuracy = " num2str(accuracy*100) "%"], "")];
title(titleString);

% save the model
save('typeIdentificationModelV2.mat', 'treeModel');

% make 2 beep sounds to signify end of operation
beep
pause(2)
beep

```

APPENDIX E: CODE FOR BUILDING FAULT LOCATION MODEL

Code for building model to locate where fault has occurred.

```

% load the full dataset
fullDataset = readtable("Datasets\full_dataset.csv");

% define the predictors and the response data
predictorData = fullDataset(fullDataset.faultIndicator == 1,5:end);
responseData = fullDataset(fullDataset.faultIndicator == 1,{ 'faultLocation' });

% split data into training and test sets
cv = cvpartition(size(predictorData, 1), 'HoldOut', 0.2);
trainIndex = training(cv);
testIndex = test(cv);

trainPredictorData = predictorData(trainIndex, :);
trainResponseData = responseData(trainIndex, :);
testPredictorData = predictorData(testIndex, :);
testResponseData = responseData(testIndex, :);

% train a decision tree model on the dataset
treeModel = fitrtree(trainPredictorData, trainResponseData);

% check the model performance
modelLoss = loss(treeModel, testPredictorData, testResponseData);
disp(['Model MSE: ', num2str(modelLoss)]);

% save the model
save('faultLocationModel.mat', 'treeModel');

% make 2 beep sounds to signify end of operation
beep
pause(2)
beep

```

APPENDIX F: CODE FOR GETTING FAULT DETECTION TIMES

The code below was used to get the detection times of faults which were reported in chapter 4.

```

% specify the transmission lines in the transmission network
faultLineArray = ["7-8" "8-9" "9-6" "6-4" "4-5" "5-7"];

```

```

% specify types of faults
faultTypeArray = ["ABCG" "ABC" "ABG" "ACG" "BCG" "AG" "BG" "CG" "AB" "AC" "BC"];

% set fault duration
faultDuration = '[4/60 11/60]';

% open the Simulink model
open_system('IEEE_9bus_new_o');

for faultLine = faultLineArray
    % switch off all faults for all transmission lines
    faultBlock = strjoin(["IEEE_9bus_new_o/Three-Phase Fault L" faultLine], "");
    set_param(faultBlock, 'FaultA', 'off');
    set_param(faultBlock, 'FaultB', 'off');
    set_param(faultBlock, 'FaultC', 'off');
    set_param(faultBlock, 'GroundFault', 'off');

    % set the fault locations to 50km
    eval(strjoin(["L" strrep(faultLine, "-", "") "_Length = 50" ";"], ""));
end

% for each transmission line
for faultLine = faultLineArray
    % select fault block for current transmission line
    faultBlock = strjoin(["IEEE_9bus_new_o/Three-Phase Fault L" faultLine], "");
    % set duration of fault for this block
    set_param(faultBlock, 'SwitchTimes', faultDuration)

    % for each type of fault
    for faultType = faultTypeArray
        % set the type of fault to the current one
        switch faultType
            case "ABCG"
                set_param(faultBlock, 'FaultA', 'on');
                set_param(faultBlock, 'FaultB', 'on');
                set_param(faultBlock, 'FaultC', 'on');
                set_param(faultBlock, 'GroundFault', 'on');
            case "ABC"
                set_param(faultBlock, 'FaultA', 'on');
                set_param(faultBlock, 'FaultB', 'on');

```

```

    set_param(faultBlock, 'FaultC', 'on');
    set_param(faultBlock, 'GroundFault', 'off');
case "ABG"
    set_param(faultBlock, 'FaultA', 'on');
    set_param(faultBlock, 'FaultB', 'on');
    set_param(faultBlock, 'FaultC', 'off');
    set_param(faultBlock, 'GroundFault', 'on');
case "ACG"
    set_param(faultBlock, 'FaultA', 'on');
    set_param(faultBlock, 'FaultB', 'off');
    set_param(faultBlock, 'FaultC', 'on');
    set_param(faultBlock, 'GroundFault', 'on');
case "BCG"
    set_param(faultBlock, 'FaultA', 'off');
    set_param(faultBlock, 'FaultB', 'on');
    set_param(faultBlock, 'FaultC', 'on');
    set_param(faultBlock, 'GroundFault', 'on');
case "AG"
    set_param(faultBlock, 'FaultA', 'on');
    set_param(faultBlock, 'FaultB', 'off');
    set_param(faultBlock, 'FaultC', 'off');
    set_param(faultBlock, 'GroundFault', 'on');
case "BG"
    set_param(faultBlock, 'FaultA', 'off');
    set_param(faultBlock, 'FaultB', 'on');
    set_param(faultBlock, 'FaultC', 'off');
    set_param(faultBlock, 'GroundFault', 'on');
case "CG"
    set_param(faultBlock, 'FaultA', 'off');
    set_param(faultBlock, 'FaultB', 'off');
    set_param(faultBlock, 'FaultC', 'on');
    set_param(faultBlock, 'GroundFault', 'on');
case "AB"
    set_param(faultBlock, 'FaultA', 'on');
    set_param(faultBlock, 'FaultB', 'on');
    set_param(faultBlock, 'FaultC', 'off');
    set_param(faultBlock, 'GroundFault', 'off');
case "AC"
    set_param(faultBlock, 'FaultA', 'on');
    set_param(faultBlock, 'FaultB', 'off');

```

```

        set_param(faultBlock, 'FaultC', 'on');
        set_param(faultBlock, 'GroundFault', 'off');
    case "BC"
        set_param(faultBlock, 'FaultA', 'off');
        set_param(faultBlock, 'FaultB', 'on');
        set_param(faultBlock, 'FaultC', 'on');
        set_param(faultBlock, 'GroundFault', 'off');
    end

    % simulate the model with the modifications
    sim('IEEE_9bus_new_o.slx');

    % save the generated data for the current configurations to a
    % csv file
    saveToCSV2(faultDetectionData, ...
        strjoin(["Datasets/Line" strrep(faultLine, "-", "") ...
            "_" string(faultType) "_Fault_Detection.csv"], ""));

    disp(strjoin(["Done generating data for Line" faultLine ...
        "and type" faultType]))
end

% switch off all faults for current transmission line
set_param(faultBlock, 'FaultA', 'off');
set_param(faultBlock, 'FaultB', 'off');
set_param(faultBlock, 'FaultC', 'off');
set_param(faultBlock, 'GroundFault', 'off');

% display message to show end of data generation for current line
disp(repmat('=',1,44))
disp(strjoin(["Done generating data for Line" faultLine]))
disp(repmat('=',1,44))
end

% make 2 beep sounds to signify end of operation
beep
pause(2)
beep

```